STAT 4360
Statistical Learning
Lecture Notes

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Contents

1 Statistical Learning and the Regression Function **1**

1.1 Data, notation, and learning tasks 1

1.2 A model for a quantitative response 1

1.3 Why the conditional mean is the regression target 2

2 Learning Error, Bias–Variance, and Nearest Neighbors **3**

2.1 Prediction and inference 3

2.2 Estimators, sampling distributions, and mean squared error 3

2.3 Reducible and irreducible prediction error 4

2.4 Parametric and nonparametric methods 5

2.5 Training error, test error, and flexibility 5

2.6 Classification and the Bayes rule 6

2.7 Nearest-neighbor classification and regression 7

3 Linear Regression: Estimation and Inference **8**

3.1 What makes a model linear? 8

3.2 Matrix notation and least squares 9

3.3 Residual variation and degrees of freedom 10

3.4 Explained variation and the ANOVA identity 10

3.5 Normal errors and inference 11

3.6 Interactions between predictors 13

3.7 Categorical predictors 14

3.8 Confidence intervals and prediction intervals 14

3.9 Leverage and the fitted residuals 15

3.10 Issues to check in a linear model 16

4 Generative Classification and Discriminant Analysis **17**

4.1 Prior probabilities, class-conditional distributions, and posteriors 17

4.2 Random vectors and the multivariate normal distribution 18

4.3 Discriminant functions 19

4.4 Linear discriminant analysis . . . 19
4.5 Quadratic discriminant analysis . . . 21
4.6 Naive Bayes . . . 22
4.7 Measuring classification performance . . . 23
4.8 ROC curves and area under the curve . . . 24

5 Logistic Regression and Direct Probability Models 27
5.1 Link functions for a binary response . . . 27
5.2 Interpreting coefficients through odds . . . 28
5.3 Fitting and the decision boundary . . . 28
5.4 Multinomial logistic regression and softmax . . . 30
5.5 Analytical comparison with discriminant analysis . . . 31

6 Generalized Linear Models and Likelihood Inference 32
6.1 The three components of a model . . . 32
6.2 Exponential-family form . . . 33
6.3 Maximum likelihood and coefficient uncertainty . . . 33
6.4 Null models, saturated models, and deviance . . . 35
6.5 Information criteria . . . 36

7 Cross-Validation and Prediction Error 37
7.1 Resampling and the validation-set approach . . . 37
7.2 Leave-one-out cross-validation . . . 37
7.3 k-fold cross-validation . . . 39
7.4 Cross-validation for classification . . . 39

8 Model Selection, Regularization, and Dimension Reduction 41
8.1 Selecting a Subset of Predictors . . . 41
8.2 Comparing Models of Different Sizes . . . 43
8.3 Shrinkage and Regularization . . . 44
8.4 Principal Components and Dimension Reduction . . . 47

9 Beyond Linearity 50
9.1 Expanding the Predictor Space . . . 50
9.2 Regression Splines . . . 51
9.3 Smoothing Splines . . . 52
9.4 Local Regression . . . 53
9.5 Generalized Additive Models . . . 54

10 Tree-Based Models **55**

10.1 Regression Trees and Terminal Regions 55

10.2 Recursive Binary Splitting 56

10.3 Classification Trees 57

11 Maximum-Margin Classification **58**

11.1 Separating Hyperplanes and the Margin 58

11.2 Equivalent Optimization Formulations 59

11.3 Lagrange Multipliers and Support Vectors 60

12 Clustering **61**

12.1 Unsupervised Learning and Dissimilarity 61

12.2 Within- and Between-Cluster Sums of Squares 61

12.3 The K-Means Algorithm 62

12.4 Agglomerative Hierarchical Clustering 63

12.5 Choosing a Dissimilarity Measure 66

Chapter 1

Statistical Learning and the Regression Function

1.1 Data, notation, and learning tasks

Let Y denote an output or response variable, and let $X = (X_1, \dots, X_p)^T$ denote the input variables, predictors, or covariates. We observe data

$$(Y_i, X_i), \qquad i = 1, \dots, n, \qquad X_i = (X_{i1}, \dots, X_{ip})^T.$$

The index i labels a subject or observation, and $j = 1, \dots, p$ labels a variable. Statistical learning studies the relationship between X and Y. We may want to understand this relationship for inference, or use X to predict Y.

Definition 1.1 (Supervised and unsupervised learning)**.** In supervised learning, both the inputs X_i and the corresponding responses Y_i are observed. In unsupervised learning, only the inputs X_i are observed; we seek structure in these inputs, for example through dimension reduction. There is no observed response to serve as a prediction target in the training data.

Definition 1.2 (Regression and classification)**.** A regression problem has a quantitative response. A classification problem has a categorical response, with the objective of assigning an observation to a class or estimating its class probabilities. Predictors may be quantitative or categorical in either problem.

1.2 A model for a quantitative response

Suppose that

$$Y = f(X) + \varepsilon. \tag{1.1}$$

The function f captures the systematic information that X provides about Y. It is fixed but unknown. The error ε represents random variability and effects of variables not included among the predictors.

Definition 1.3 (Regression function)**.** For an integrable response, the regression function is

$$f(x) = \mathbb{E}(Y \mid X = x).$$

Here x denotes the fixed predictor value at which the conditional mean is evaluated.

The usual mean-zero, constant-variance error assumptions are

$$\mathbb{E}(\varepsilon \mid X) = 0, \qquad \mathrm{Var}(\varepsilon \mid X) = \sigma^2. \tag{1.2}$$

A sufficient assumption is that ε is independent of X, with $\mathbb{E}(\varepsilon) = 0$ and $\mathrm{Var}(\varepsilon) = \sigma^2$.

Proposition 1.4 (Conditional mean and variance)**.** *Under* (1.1)–(1.2)*,*

$$\mathbb{E}(Y \mid X) = f(X), \qquad \mathrm{Var}(Y \mid X) = \sigma^2.$$

Proof. Because $f(X)$ is fixed conditional on X,

$$\begin{aligned}\mathbb{E}(Y \mid X) &= \mathbb{E}\{f(X) + \varepsilon \mid X\} \\ &= \mathbb{E}\{f(X) \mid X\} + \mathbb{E}(\varepsilon \mid X) \\ &= f(X) + 0 = f(X).\end{aligned}$$

Similarly,

$$\mathrm{Var}(Y \mid X) = \mathrm{Var}\{f(X) + \varepsilon \mid X\} = \mathrm{Var}(\varepsilon \mid X) = \sigma^2.$$

Note 1.5. If $f(X) = \mathbb{E}(Y \mid X)$ and $\varepsilon = Y - f(X)$, the conditional mean-zero property follows from the definition. Independence and constant conditional variance are additional assumptions; they do not follow from the definition of the regression function.

1.3 Why the conditional mean is the regression target

For squared-error prediction, the target is not an arbitrary description of Y: it is the function of X that minimizes expected prediction error. Assume $\mathbb{E}(Y^2) < \infty$, and write $m(x) = \mathbb{E}(Y \mid X = x)$.

Exercise 1.1. Use conditioning to show that $m(X)$ is a best predictor of Y under squared error. At a fixed x, minimize

$$G_x(c) = \mathbb{E}\{(Y - c)^2 \mid X = x\}$$

over constants c.

Solution. Add and subtract the conditional mean:

$$\begin{aligned}G_x(c) &= \mathbb{E}\{[Y - m(x) + m(x) - c]^2 \mid X = x\} \\ &= \mathbb{E}\{[Y - m(x)]^2 \mid X = x\} + [m(x) - c]^2 \\ &\quad + 2[m(x) - c]\mathbb{E}\{Y - m(x) \mid X = x\} \\ &= \mathrm{Var}(Y \mid X = x) + [m(x) - c]^2.\end{aligned}$$

The cross term is zero. Thus $c = m(x)$ minimizes $G_x(c)$. For any measurable predictor g with finite squared error, iterated expectation gives

$$\begin{aligned}\mathbb{E}\{[Y - g(X)]^2\} &= \mathbb{E}\{\mathrm{Var}(Y \mid X)\} + \mathbb{E}\{[m(X) - g(X)]^2\} \\ &\geq \mathbb{E}\{\mathrm{Var}(Y \mid X)\}.\end{aligned}$$

Equality holds precisely when $g(X) = m(X)$ almost surely. In particular, if $Y = f(X) + \varepsilon$ and $\mathbb{E}(\varepsilon \mid X) = 0$, then $m(X) = f(X)$. When $\mathrm{Var}(\varepsilon \mid X) = \sigma^2$, the minimum prediction error is σ^2, the irreducible error. Its sources can include unmeasured predictors and other variation not determined by the observed X.

Chapter 2

Learning Error, Bias–Variance, and Nearest Neighbors

2.1 Prediction and inference

Suppose that the observations satisfy

$$Y_i = f(X_i) + \varepsilon_i, \qquad i = 1, \dots, n.$$

We assume that, conditional on the predictors, the errors have mean zero, variance σ^2, and are mutually independent. Thus

$$\mathbb{E}(Y_i \mid X_i) = f(X_i), \qquad \mathrm{Var}(Y_i \mid X_i) = \sigma^2.$$

Under independent sampling, the responses are also conditionally independent given the predictors. Learning means using the data to obtain an estimate $\hat{f}$ of f.

For *prediction*, we use $\hat{Y} = \hat{f}(x)$ at a new predictor value x. The fitted function may be treated as a black box: its exact form and the interpretation of its components need not be the main concern. For *inference*, we ask which predictors are associated with Y and how Y changes with the components of X. The form of the model and the meaning of its parameters matter. Linear models and their extensions are useful for inference; highly nonlinear methods may be useful for prediction. The intended use affects the choice of method.

2.2 Estimators, sampling distributions, and mean squared error

Suppose that θ is an unknown target and $\hat{\theta}$ is an estimator computed from data. Before observing the data, $\hat{\theta}$ is a random quantity. Its probability distribution is called its *sampling distribution*.

Example 2.1 (Estimating a population mean)**.** Let $X_1, \dots, X_n$ be independent and identically distributed with $\mathbb{E}(X_i) = \theta$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Two possible estimators are $\bar{X} = n^{-1} \sum_i X_i$ and X_1. Both are unbiased, but

$$\mathbb{E}\big[(\bar{X} - \theta)^2\big] = \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}, \qquad \mathbb{E}\big[(X_1 - \theta)^2\big] = \sigma^2.$$

The sample mean uses all the observations and has smaller mean squared error when $n > 1$ and $\sigma^2 > 0$. The law of large numbers gives $\bar{X} \longrightarrow \mathbb{E}(X_1)$ in probability as $n \to \infty$.

Definition 2.2. For an estimator with finite second moment, define

$$\begin{aligned}\mathrm{Bias}(\hat{\theta}) &= \mathbb{E}(\hat{\theta}) - \theta,\\ \mathrm{MSE}(\hat{\theta}) &= \mathbb{E}\big[(\hat{\theta} - \theta)^2\big],\\ \mathrm{Var}(\hat{\theta}) &= \mathbb{E}\big[(\hat{\theta} - \mathbb{E}\hat{\theta})^2\big].\end{aligned}$$

Bias measures the difference between the estimator's average value and its target. Variance measures its average squared fluctuation around its own mean.

Proposition 2.3 (Bias–variance decomposition)**.**

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Bias}(\hat{\theta})^2 + \mathrm{Var}(\hat{\theta}).$$

Proof. Add and subtract $\mathbb{E}\hat{\theta}$:

$$\begin{aligned}\mathbb{E}\big[(\hat{\theta} - \theta)^2\big] &= \mathbb{E}\big[(\hat{\theta} - \mathbb{E}\hat{\theta} + \mathbb{E}\hat{\theta} - \theta)^2\big]\\ &= \mathbb{E}\big[(\hat{\theta} - \mathbb{E}\hat{\theta})^2\big] + (\mathbb{E}\hat{\theta} - \theta)^2\\ &\quad + 2(\mathbb{E}\hat{\theta} - \theta)\mathbb{E}(\hat{\theta} - \mathbb{E}\hat{\theta}).\end{aligned}$$

The last expectation is zero, and $\mathbb{E}\hat{\theta} - \theta$ is a constant. The remaining two terms are the variance and squared bias.

An unbiased estimator need not be the best estimator under squared error. A biased estimator can have a sufficiently smaller variance to give smaller MSE.

2.3 Reducible and irreducible prediction error

Let x_0 be a predictor value at which we want to predict a new response $Y_0 = f(x_0) + \varepsilon_0$. The test observation is independent of the training data. Assume $\mathbb{E}(\varepsilon_0 \mid x_0) = 0$ and $\mathrm{Var}(\varepsilon_0 \mid x_0) = \sigma^2$. The prediction is $\hat{Y}_0 = \hat{f}(x_0)$.

The estimation error is $\hat{f}(x_0) - f(x_0)$, whereas the prediction error is $\hat{Y}_0 - Y_0$. At this fixed x_0, define

$$\mathrm{Bias}\{\hat{f}(x_0)\} = \mathbb{E}\{\hat{f}(x_0)\} - f(x_0).$$

The expectation here is over repeated training samples. Bias can arise, for example, from approximating a complicated true function with a simpler model. The preceding decomposition gives

$$\mathbb{E}\big[\{\hat{f}(x_0) - f(x_0)\}^2\big] = \mathrm{Bias}\{\hat{f}(x_0)\}^2 + \mathrm{Var}\{\hat{f}(x_0)\}.$$

Exercise 2.1. Show that the mean squared prediction error at x_0 is

$$\mathbb{E}\big[(\hat{Y}_0 - Y_0)^2 \mid X_0 = x_0\big] = \mathrm{Bias}\{\hat{f}(x_0)\}^2 + \mathrm{Var}\{\hat{f}(x_0)\} + \sigma^2.$$

Solution. Write

$$\hat{Y}_0 - Y_0 = \hat{f}(x_0) - f(x_0) - \varepsilon_0.$$

After squaring and taking expectations, the cross term vanishes because the new error is independent of the training data and has mean zero:

$$\begin{aligned}\mathbb{E}[(\hat{Y}_0 - Y_0)^2 \mid X_0 = x_0] &= \mathbb{E}[\{\hat{f}(x_0) - f(x_0)\}^2] + \mathbb{E}(\varepsilon_0^2 \mid x_0) \\ &= \mathrm{Bias}\{\hat{f}(x_0)\}^2 + \mathrm{Var}\{\hat{f}(x_0)\} + \sigma^2.\end{aligned}$$

The first two terms form the *reducible error*: a better estimate of f can reduce them. The last term is the *irreducible error*, the variation in a new response that remains even when f is known. It supplies the lower bound σ^2 under these assumptions.

2.4 Parametric and nonparametric methods

A *parametric* method assumes a functional form and estimates its parameters. For example,

$$f(x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p.$$

This reduces estimation to a finite number of coefficients and often makes fitting and interpretation easier. The assumed form may only approximate the true unknown function. More flexible parametric families can approximate more shapes, but estimating more parameters can lead to overfitting.

A *nonparametric* method does not prescribe a fixed finite-dimensional functional form for f. It can fit a wide range of shapes, but it does not reduce the problem to estimating a small fixed number of parameters. Nonparametric methods still require assumptions, such as sufficient local regularity to learn from nearby observations.

2.5 Training error, test error, and flexibility

For training data (X_i, Y_i), $i = 1, \ldots, n$, the training mean squared error is

$$\mathrm{MSE}_{\mathrm{train}} = \frac{1}{n} \sum_{i=1}^{n} \{Y_i - \hat{f}(X_i)\}^2.$$

Least squares chooses a function from a candidate family $\mathcal{F}$:

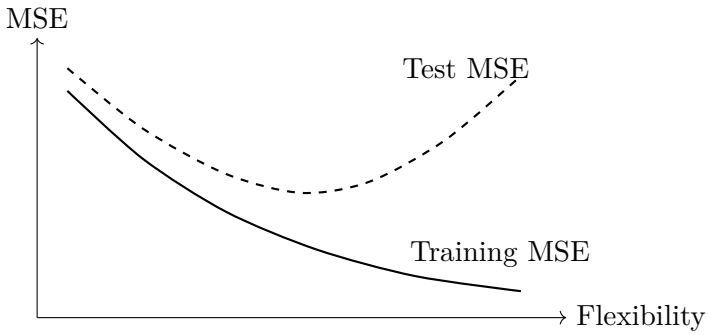
$$\hat{f} \in \arg\min_{g \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \{Y_i - g(X_i)\}^2.$$

The model should also work well on observations that were not used for fitting. For an independent test set of size m, the test MSE is

$$\mathrm{MSE}_{\mathrm{test}} = \frac{1}{m} \sum_{j=1}^{m} \{Y_j^{\mathrm{test}} - \hat{f}(X_j^{\mathrm{test}})\}^2.$$

A small training MSE does not guarantee a small test MSE.

For nested candidate families, minimizing training squared error over a larger family cannot increase the minimum training MSE. Test error often has a U-shaped pattern: increased flexibility first improves approximation, but excessive flexibility increases sensitivity to the particular training sample. A model can follow noise and individual training points closely while predicting poorly on new observations. This is *overfitting*.



Typical schematic: test MSE need not be U-shaped for every problem.

The bias–variance decomposition explains the trade-off. As flexibility increases, bias usually decreases and variance usually increases. Too much flexibility can make the variance dominate. The challenge is to achieve both small bias and small variance, rather than to minimize training error alone. Accuracy for a quantitative response is assessed by test MSE. Cross-validation uses the available training data to estimate predictive error and select flexibility; an untouched test set can then assess the selected procedure.

2.6 Classification and the Bayes rule

For a qualitative response, let Y take one of the unordered class labels $1, \dots, K$. The additive numerical model $Y = f(X) + \varepsilon$ is not needed for such labels. A classifier g predicts the class $g(x)$ at a predictor value x. Its zero–one loss is

$$\mathbf{1}\{g(X) \neq Y\} = \begin{cases} 1, & g(X) \neq Y, \\ 0, & g(X) = Y. \end{cases}$$

The mean error rate is

$$\mathbb{E}\big[\mathbf{1}\{g(X) \neq Y\}\big] = \mathbb{P}\{g(X) \neq Y\}.$$

For m evaluated observations, the empirical error rate is $m^{-1} \sum_{i=1}^{m} \mathbf{1}\{\hat{Y}_i \neq Y_i\}$; the empirical accuracy is one minus this rate.

Proposition 2.4 (Bayes classifier)**.** *Set $P_k(x) = \mathbb{P}(Y = k \mid X = x)$. A classifier minimizing mean zero–one loss assigns x to any class attaining $\max_k P_k(x)$. Its error rate is*

$$R^* = 1 - \mathbb{E}\left[\max_{1 \leq k \leq K} P_k(X)\right].$$

Proof. By iterated expectation,

$$\mathbb{E}\big[\mathbf{1}\{g(X) \neq Y\}\big] = \mathbb{E}\left[\mathbb{E}\big[\mathbf{1}\{g(X) \neq Y\} \mid X\big]\right].$$

At a fixed x, choosing class c gives conditional error

$$G_c(x) = \mathbb{P}(Y \neq c \mid X = x) = 1 - P_c(x).$$

Minimizing this quantity amounts to maximizing $P_c(x)$. The minimum conditional error is $1 - \max_k P_k(x)$; averaging over X proves the result.

For example, posterior probabilities 0.3 and 0.7 for classes A and B lead to the prediction of class B.

The Bayes error rate is a lower bound for classification error under zero–one loss. The posterior class probabilities play the role that $\mathbb{E}(Y \mid X = x)$ plays in squared-error regression. In practice, these probabilities are unknown and must be estimated.

For two classes, predict class 1 if $P_1(x) > 1/2$ and class 2 otherwise, with any fixed convention at ties. The Bayes decision boundary is

$$\{x : P_1(x) = 1/2\}.$$

With two predictors this is the level set $\{(x_1, x_2) : P_1(x_1, x_2) = 1/2\}$, which need not be a straight line.

2.7 Nearest-neighbor classification and regression

Given a point x_0, choose a positive integer k and identify the indices $\mathcal{N}_0$ of the k training observations closest to x_0. Euclidean distance is one possible measure of closeness:

$$d(x, x_0) = \left\{ \sum_{j=1}^{p} (x_j - x_{0j})^2 \right\}^{1/2}.$$

Fix the distance rule when comparing different values of k.

The k-nearest-neighbor classifier estimates

$$\hat{P}_c(x_0) = \frac{1}{k} \sum_{i \in \mathcal{N}_0} \mathbf{1}\{Y_i = c\}$$

and assigns x_0 to the class with the largest estimated probability. Thus it uses a majority vote among the neighbors. A tie may be resolved by a fixed rule or by random selection among tied classes. This is a local method: only a portion of the training observations contributes at a particular x_0. Linear regression, in contrast, fits a global relationship.

For a quantitative response, k-nearest-neighbor regression estimates the conditional mean by the average response among the neighbors:

$$\hat{f}(x_0) = \frac{1}{k} \sum_{i \in \mathcal{N}_0} Y_i.$$

Both procedures are nonparametric. The number of neighbors controls flexibility and hence the bias–variance trade-off. A small k gives a flexible fit, typically with low bias and high variance. Increasing k smooths the fit, usually increasing bias and decreasing variance. With distinct training predictor values, $k = 1$ classifies every training point correctly when the point itself is allowed as a neighbor. Duplicate predictors with conflicting labels can prevent zero training error.

To select k, compare estimated prediction errors, for example by cross-validation. Plotting error against $1/k$ puts increasing flexibility from left to right. The tuning parameter k is a *hyperparameter*.

Chapter 3

Linear Regression: Estimation and Inference

3.1 What makes a model linear?

In a linear regression model, *linear* refers to the unknown coefficients, not necessarily to the original predictors. For example,

$$\beta_0 + \beta_1 x + \beta_2 x^2, \qquad \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

are linear models. The function $\beta_0 + (1 + \beta_1 x_1)^{-1}$ is nonlinear in its coefficients. The transformed response model

$$\mathbb{E}(\log Y \mid X = x) = \beta_0 + \beta_1 \log x$$

is linear in its coefficients, assuming $Y > 0$ and $x > 0$.

At the population level, a squared-error target within a candidate family $\mathcal{F}$ is

$$f^*_{\mathcal{F}} \in \arg\min_{g \in \mathcal{F}} \mathbb{E}\{g(X) - Y\}^2.$$

Its sample counterpart minimizes the empirical training MSE. These two minimizations distinguish the best function in a family from its estimate based on a finite sample.

For multiple linear regression,

$$f(x) = \mathbb{E}(Y \mid X = x) = \beta_0 + \sum_{j=1}^{p} \beta_j x_j.$$

The intercept is $\beta_0 = f(0)$, when the predictor value zero is meaningful. Holding the other predictors fixed,

$$f(x_1, \ldots, x_j + 1, \ldots, x_p) - f(x_1, \ldots, x_j, \ldots, x_p) = \beta_j.$$

Thus β_j is the change in the conditional mean associated with a one-unit increase in x_j, adjusting for the remaining predictors.

3.2 Matrix notation and least squares

For observations $(Y_i, X_{i1}, \dots, X_{ip})$, write

$$Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \qquad X = \begin{pmatrix} 1 & X_{11} & \cdots & X_{1p} \\ \vdots & \vdots & & \vdots \\ 1 & X_{n1} & \cdots & X_{np} \end{pmatrix}, \qquad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}.$$

The model is

$$Y = X\beta + \varepsilon, \qquad \mathbb{E}(\varepsilon \mid X) = 0, \qquad \operatorname{Var}(\varepsilon \mid X) = \sigma^2 I_n.$$

We treat X as fixed, or work conditionally on it. Assume $n > p + 1$ and that X has full column rank, so $(X^T X)^{-1}$ exists.

The fitted response vector and the residual vector are

$$\hat{Y} = X\hat{\beta}, \qquad e = Y - \hat{Y}.$$

For a new predictor vector $x_0 = (1, x_{01}, \dots, x_{0p})^T$, the fitted mean is $\hat{f}(x_0) = x_0^T \hat{\beta}$.

Least squares minimizes the training MSE, or equivalently the residual sum of squares:

$$\hat{\beta} = \arg\min_{b \in \mathbb{R}^{p+1}} (Y - Xb)^T (Y - Xb).$$

Expanding gives

$$(Y - Xb)^T (Y - Xb) = Y^T Y - 2Y^T Xb + b^T X^T Xb.$$

Differentiating with respect to b gives the normal equations

$$2X^T X\hat{\beta} - 2X^T Y = 0.$$

The Hessian $2X^T X$ is positive definite, so the unique minimizer is

$$\hat{\beta} = (X^T X)^{-1} X^T Y.$$

The minimized sum of squared residuals is denoted by $\mathrm{SS}_{\mathrm{ERR}} = e^T e$.

Proposition 3.1 (Mean and covariance of least squares)**.** *Under the stated mean, covariance, and rank assumptions,*

$$\mathbb{E}(\hat{\beta} \mid X) = \beta, \qquad \operatorname{Var}(\hat{\beta} \mid X) = \sigma^2 (X^T X)^{-1}.$$

Proof. Let $A = (X^T X)^{-1} X^T$. Then $\hat{\beta} = AY$ is linear in Y, and

$$\mathbb{E}(\hat{\beta} \mid X) = A\mathbb{E}(Y \mid X) = AX\beta = \beta.$$

Also,

$$\begin{aligned} \operatorname{Var}(\hat{\beta} \mid X) &= A \operatorname{Var}(Y \mid X) A^T \\ &= \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1} = \sigma^2 (X^T X)^{-1}. \end{aligned}$$

In particular, $\operatorname{Var}(\hat{\beta}_j \mid X)$ is σ^2 times the $(j+1)$st diagonal element of $(X^T X)^{-1}$.

3.2.1 Simple linear regression

With one nonconstant predictor, the fitted model is $\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x$. The normal equations yield

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

The intercept represents $\mathbb{E}(Y \mid X = 0)$ under the model. Both coefficient estimators are unbiased under the linear mean model and the zero-mean error assumption.

3.3 Residual variation and degrees of freedom

An unbiased estimator of the error variance is

$$\hat{\sigma}^2 = \frac{\mathrm{SS}_{\mathrm{ERR}}}{n - p - 1},$$

and $\hat{\sigma}$ is the *residual standard error*. It measures residual variation in the same units as Y. For simple linear regression, the denominator is $n - 2$.

Residual degrees of freedom equal the number of observations minus the number of linearly independent fitted coefficients. The same idea gives the usual sample variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2 :$$

one degree of freedom is used to estimate the mean.

Exercise 3.1. Prove that $\mathbb{E}(\hat{\sigma}^2 \mid X) = \sigma^2$.

Solution. Define the projection matrix $H = X(X^T X)^{-1} X^T$. It is symmetric and idempotent, has rank $p + 1$, and satisfies $(I - H)X = 0$. Hence

$$e = (I - H)Y = (I - H)\varepsilon, \qquad \mathrm{SS}_{\mathrm{ERR}} = \varepsilon^T (I - H)\varepsilon.$$

For a zero-mean random vector with covariance $\sigma^2 I_n$,

$$\mathbb{E}\{\varepsilon^T (I - H)\varepsilon \mid X\} = \sigma^2 \operatorname{tr}(I - H) = \sigma^2 (n - p - 1).$$

Dividing by the residual degrees of freedom proves the claim. Normality is not needed for this unbiasedness result. The square root $\hat{\sigma}$ itself need not be unbiased for σ.

3.4 Explained variation and the ANOVA identity

Assume that the model contains an intercept. Define

$$\mathrm{SS}_{\mathrm{TOT}} = \sum_i (Y_i - \bar{Y})^2 = (Y - \bar{Y}\mathbf{1})^T (Y - \bar{Y}\mathbf{1}),$$
$$\mathrm{SS}_{\mathrm{REG}} = \sum_i (\hat{Y}_i - \bar{Y})^2 = (\hat{Y} - \bar{Y}\mathbf{1})^T (\hat{Y} - \bar{Y}\mathbf{1}),$$
$$\mathrm{SS}_{\mathrm{ERR}} = \sum_i (Y_i - \hat{Y}_i)^2 = (Y - \hat{Y})^T (Y - \hat{Y}).$$

Here $\mathrm{SS}_{\mathrm{TOT}} = (n-1)s_Y^2$ is total response variation, $\mathrm{SS}_{\mathrm{REG}}$ is variation explained by the fitted regression, and $\mathrm{SS}_{\mathrm{ERR}}$ is unexplained variation. Measures of absolute residual size, such as $\hat{\sigma}$, depend on the response scale; explained variation gives a relative assessment of fit, useful when observations differ in scale, size, or other characteristics.

Exercise 3.2. Establish the identity $\mathrm{SS}_{\mathrm{TOT}} = \mathrm{SS}_{\mathrm{REG}} + \mathrm{SS}_{\mathrm{ERR}}$.

Solution. For each observation,

$$Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + (Y_i - \hat{Y}_i).$$

The least-squares normal equations imply $X^T e = 0$. Because the constant vector and $\hat{Y}$ belong to the column space of X,

$$\sum_i e_i = 0, \qquad \sum_i \hat{Y}_i e_i = 0, \qquad \sum_i (\hat{Y}_i - \bar{Y}) e_i = 0.$$

Squaring the first decomposition and summing therefore gives

$$\begin{aligned}\sum_i (Y_i - \bar{Y})^2 &= \sum_i (\hat{Y}_i - \bar{Y})^2 + \sum_i e_i^2 + 2\sum_i (\hat{Y}_i - \bar{Y}) e_i \\ &= \mathrm{SS}_{\mathrm{REG}} + \mathrm{SS}_{\mathrm{ERR}}.\end{aligned}$$

If $\mathrm{SS}_{\mathrm{TOT}} > 0$, the coefficient of determination is

$$R^2 = \frac{\mathrm{SS}_{\mathrm{REG}}}{\mathrm{SS}_{\mathrm{TOT}}} = 1 - \frac{\mathrm{SS}_{\mathrm{ERR}}}{\mathrm{SS}_{\mathrm{TOT}}}, \qquad 0 \le R^2 \le 1.$$

It is the proportion of total response variation explained by the regression. A large value indicates a close fit to the training responses. When $\hat{Y}$ is nonconstant, R^2 is also the square of the sample correlation between Y and $\hat{Y}$. In simple linear regression, it is the square of the sample correlation between X and Y: a large absolute correlation means the points are closely clustered about a line.

Adding predictors cannot decrease R^2, since the original fitted model remains available in the larger model. The adjusted coefficient of determination accounts for residual degrees of freedom:

$$R^2_{\mathrm{adj}} = 1 - \frac{\mathrm{SS}_{\mathrm{ERR}}/(n-p-1)}{\mathrm{SS}_{\mathrm{TOT}}/(n-1)} = 1 - \frac{\hat{\sigma}^2}{s_Y^2}.$$

It rewards an additional predictor only when the reduction in residual sum of squares is sufficient to offset the loss of a degree of freedom. Merely reducing the residual sum of squares is not sufficient.

3.5 Normal errors and inference

For exact finite-sample normal-theory inference, assume $\sigma^2 > 0$ and

$$\varepsilon \mid X \sim N_n(0, \sigma^2 I_n), \qquad Y \mid X \sim N_n(X\beta, \sigma^2 I_n).$$

Then

$$\hat{\beta} \mid X \sim N_{p+1}(\beta, \sigma^2 (X^T X)^{-1}), \qquad \frac{(n-p-1)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-p-1}.$$

Moreover, $\hat{\beta}$ and $\hat{\sigma}^2$ are independent. In particular, each coefficient estimator has a univariate normal distribution.

Exercise 3.3. Explain the chi-square distribution and independence stated above, and show that the least-squares coefficient estimators are maximum likelihood estimators under normal errors.

Solution. Write $Z = \varepsilon/\sigma \sim N_n(0, I_n)$. In an orthonormal basis adapted to the column space of X, the projection $(I - H)Z$ has $n - p - 1$ independent standard normal coordinates. Thus

$$\frac{\mathrm{SS}_{\mathrm{ERR}}}{\sigma^2} = Z^T(I - H)Z \sim \chi^2_{n-p-1}.$$

The fitted coefficients depend on the projection $H\varepsilon$, whereas the residuals depend on $(I - H)\varepsilon$. These projections have covariance $\sigma^2 H(I - H) = 0$. Joint normality makes them independent, so $\hat{\beta}$ and any function of the residuals, including $\hat{\sigma}^2$, are independent.

The normal log likelihood, up to a constant, is

$$\ell(b, \sigma^2) = -\frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}(Y - Xb)^T(Y - Xb).$$

For each fixed $\sigma^2 > 0$, maximizing this expression over b is precisely least squares. Hence $\hat{\beta}$ is the maximum likelihood estimator. When the residual sum of squares is positive, maximizing over the variance gives $\hat{\sigma}^2_{\mathrm{ML}} = \mathrm{SS}_{\mathrm{ERR}}/n$, which differs from the unbiased variance estimator.

3.5.1 Testing a particular predictor

To ask whether predictor X_j is associated with the response after adjusting for the other predictors, test

$$H_0 : \beta_j = 0 \qquad \text{against} \qquad H_1 : \beta_j \neq 0.$$

Let c_{jj} denote the $(j+1)$st diagonal element of $(X^TX)^{-1}$. Then

$$\mathrm{SE}(\hat{\beta}_j) = \hat{\sigma}\sqrt{c_{jj}}, \qquad T_j = \frac{\hat{\beta}_j}{\mathrm{SE}(\hat{\beta}_j)} \sim t_{n-p-1} \quad \text{under } H_0.$$

A two-sided level-α test rejects for $|T_j| > t_{1-\alpha/2,n-p-1}$, where $t_{u,\nu}$ denotes the u quantile of the t distribution with ν degrees of freedom. The corresponding confidence interval is

$$\hat{\beta}_j \;\pm\; t_{1-\alpha/2,n-p-1}\,\mathrm{SE}(\hat{\beta}_j).$$

3.5.2 Testing the regression as a whole

The overall significance test is

$$H_0 : \beta_1 = \cdots = \beta_p = 0 \quad \text{against} \quad H_1 : \text{at least one slope is nonzero.}$$

The ANOVA table is

Source	Sum of squares	Degrees of freedom	Mean square
Regression	$\mathrm{SS}_{\mathrm{REG}}$	p	$\mathrm{MS}_{\mathrm{REG}} = \mathrm{SS}_{\mathrm{REG}}/p$
Error	$\mathrm{SS}_{\mathrm{ERR}}$	$n - p - 1$	$\mathrm{MS}_{\mathrm{ERR}} = \hat{\sigma}^2$
Total	$\mathrm{SS}_{\mathrm{TOT}}$	$n - 1$	

Under H_0,

$$F = \frac{\mathrm{MS}_{\mathrm{REG}}}{\mathrm{MS}_{\mathrm{ERR}}} = \frac{\mathrm{SS}_{\mathrm{REG}}/p}{\hat{\sigma}^2} \sim F_{p,n-p-1}.$$

Large values give evidence against H_0.

3.5.3 Comparing nested models

A reduced model is nested within a full model if its predictor columns span a subspace of the full model's predictor space. In particular, this holds when its predictors are a subset of the full model's predictors. Let the full model have $p+1$ coefficients, and suppose the reduced model sets q of the full model's independent coefficients to zero. The extra sum of squares is

$$\begin{aligned}\mathrm{SS}_{\mathrm{EX}} &= \mathrm{SS}_{\mathrm{REG}}(\text{full}) - \mathrm{SS}_{\mathrm{REG}}(\text{reduced}) \\ &= \mathrm{SS}_{\mathrm{ERR}}(\text{reduced}) - \mathrm{SS}_{\mathrm{ERR}}(\text{full}).\end{aligned}$$

It measures the additional variation explained by the full model.

Exercise 3.4. Construct the partial F statistic for testing the q restrictions defining the reduced model, and justify its null distribution.

Solution. Let H_F and H_R be the projection matrices for the full and reduced models. Nesting implies that $H_F - H_R$ is an orthogonal projection of rank q. Consequently,

$$\mathrm{SS}_{\mathrm{EX}} = Y^T(H_F - H_R)Y.$$

Under the reduced-model null hypothesis, the mean vector lies in the reduced space. Thus $\mathrm{SS}_{\mathrm{EX}}/\sigma^2 \sim \chi^2_q$. The full-model residual sum of squares divided by σ^2 is χ^2_{n-p-1}. These two quadratic forms are independent because the corresponding normal projections are orthogonal. Therefore

$$F = \frac{\{\mathrm{SS}_{\mathrm{ERR}}(\text{reduced}) - \mathrm{SS}_{\mathrm{ERR}}(\text{full})\}/q}{\mathrm{SS}_{\mathrm{ERR}}(\text{full})/(n-p-1)} \sim F_{q,n-p-1}$$

under the null. Reject for sufficiently large F. The overall regression test is the special case in which the reduced model contains only an intercept.

3.6 Interactions between predictors

The additive model

$$f(x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

assumes that the effect of one predictor does not depend on the value of the other. A one-unit increase in x_1 changes the mean by β_1, regardless of x_2.

Include the product x_1x_2 to allow interaction:

$$\begin{aligned}f(x_1, x_2) &= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 \\ &= \beta_0 + (\beta_1 + \beta_3 x_2)x_1 + \beta_2 x_2.\end{aligned}$$

The slope for x_1 is now $\beta_1 + \beta_3 x_2$, so it depends on x_2. Likewise, the slope for x_2 is $\beta_2 + \beta_3 x_1$.

Example 3.2 (TV and radio advertising)**.** For a model with TV, radio, and their product, the estimated slope of TV is

$$\hat{\beta}_1 + \hat{\beta}_3 \,\text{radio}.$$

For example, a fitted slope $19 + 0.1\,\text{radio}$ means that a one-unit increase in TV advertising changes fitted sales by this amount, holding radio advertising fixed. If a TV unit is \$1,000, this describes a \$1,000 increase in the TV budget. The sales and radio units must be those used in fitting.

If the interaction is significant but a main effect is not, the main effect should generally be retained. With an interaction present, the main effect refers to the slope when the other predictor is zero; it does not summarize the effect at every value. Retaining the constituent main effects also preserves the usual hierarchy of the model.

3.7 Categorical predictors

A categorical predictor should be treated as a factor, rather than assigned arbitrary numerical scores. For C categories, use $C-1$ binary indicator variables together with an intercept. The omitted category is the reference.

Example 3.3 (Operating system)**.** Suppose the categories are Mac and Windows. Take Mac as the reference and let $z=1$ for Windows and $z=0$ for Mac. The model is

$$\mathbb{E}(Y \mid \text{OS}) = \beta_0 + \beta_1 z.$$

Then

$$\beta_0 = \mathbb{E}(Y \mid \text{OS} = \text{Mac}), \qquad \beta_1 = \mathbb{E}(Y \mid \text{OS} = \text{Windows}) - \mathbb{E}(Y \mid \text{OS} = \text{Mac}).$$

Example 3.4 (Three categories)**.** For the categories White, AA, and Other, use White as the reference and set

Category	$z_1 = \mathbf{1}\{\text{AA}\}$	$z_2 = \mathbf{1}\{\text{Other}\}$
AA	1	0
Other	0	1
White	0	0

In the model $\mathbb{E}(Y \mid \text{category}) = \beta_0 + \beta_1 z_1 + \beta_2 z_2$, the three means are β_0, $\beta_0+\beta_1$, and $\beta_0+\beta_2$ for White, AA, and Other, respectively.

To test the significance of a categorical variable with C categories, simultaneously test the $C-1$ associated coefficients:

$$H_0 : \beta_1 = \cdots = \beta_{C-1} = 0.$$

This is a partial F test comparing models with and without those indicators, while retaining the other predictors. A regression can include several categorical predictors, quantitative predictors, and interactions between them.

3.8 Confidence intervals and prediction intervals

A confidence interval for $f(x_0)$ concerns the unknown *mean response* at x_0. Its repeated-sampling coverage is

$$\mathbb{P}\{L \le f(x_0) \le U\} = 1-\alpha$$

under the normal linear model, with x_0 fixed. The endpoints are random; $f(x_0)$ is fixed.

Exercise 3.5. Derive a confidence interval for $f(x_0) = x_0^T\beta$ and a prediction interval for a new response $Y_0 = x_0^T\beta + \varepsilon_0$. Which interval is wider?

Solution. Put $\nu = n - p - 1$ and $h_0 = x_0^T (X^T X)^{-1} x_0$. Since

$$\mathrm{Var}\{\hat{f}(x_0) \mid X\} = \sigma^2 h_0,$$

and the standardized fitted mean is normal and independent of $\hat{\sigma}^2$, the confidence interval is

$$\hat{f}(x_0) \ \pm \ t_{1-\alpha/2,\nu} \hat{\sigma} \sqrt{h_0}.$$

For a new independent error $\varepsilon_0 \sim N(0, \sigma^2)$,

$$Y_0 - \hat{f}(x_0) = \varepsilon_0 - x_0^T (\hat{\beta} - \beta), \qquad \mathrm{Var}\{Y_0 - \hat{f}(x_0) \mid X\} = \sigma^2 (1 + h_0).$$

This prediction error is independent of the fitted residual variance, so

$$\hat{f}(x_0) \ \pm \ t_{1-\alpha/2,\nu} \hat{\sigma} \sqrt{1 + h_0}$$

is the prediction interval. Its additional 1 accounts for the noise in a new individual response, so it is wider when $\hat{\sigma} > 0$.

3.9 Leverage and the fitted residuals

Suppose that the linear-model design matrix X has $p + 1$ linearly independent columns, including the intercept. Recall that

$$\hat{Y} = HY, \qquad H = X(X^T X)^{-1} X^T, \qquad e = (I - H)Y.$$

The diagonal entry $h_i = H_{ii}$ is the *leverage* of observation i. It depends on the predictor values, not on the observed response.

Exercise 3.6. Explain the meaning of high leverage. Find the range and average of the leverages, and calculate the residual variance under $\mathrm{Var}(Y \mid X) = \sigma^2 I_n$.

Solution. The fitted response is $\hat{Y}_i = \sum_j H_{ij} Y_j$, so the coefficient of its own observed response is h_i. Equivalently, $\partial \hat{Y}_i / \partial Y_i = h_i$. A predictor value far from the main predictor cloud can have high leverage and can strongly affect its own fitted value. High leverage does not by itself mean that the response is an outlier.

Let u_i be the ith coordinate vector. Since H and $I - H$ are orthogonal projections,

$$h_i = u_i^T H u_i = \|H u_i\|^2 \geq 0, \qquad 1 - h_i = u_i^T (I - H) u_i \geq 0.$$

Hence $0 \leq h_i \leq 1$. Their sum is the rank of the projection:

$$\sum_{i=1}^{n} h_i = \mathrm{tr}(H) = p + 1, \qquad \frac{1}{n} \sum_{i=1}^{n} h_i = \frac{p+1}{n}.$$

A leverage substantially above this average deserves attention. Finally,

$$\mathrm{Var}(e \mid X) = \sigma^2 (I - H)(I - H)^T = \sigma^2 (I - H), \qquad \mathrm{Var}(e_i \mid X) = \sigma^2 (1 - h_i).$$

A high-leverage observation can therefore have a small fitted residual even when its effect on the fit is substantial.

3.10 Issues to check in a linear model

Non-normal errors do not invalidate least-squares unbiasedness or its covariance formula when the mean and covariance assumptions hold. Exact normal-theory t and F conclusions require normality. Large-sample approximations may remain useful with non-normal errors under suitable moment and design conditions; a large sample alone is not a substitute for these conditions.

If X is not full rank, coefficients are not uniquely identified without constraints or a change of parameterization. Near collinearity can produce unstable coefficient estimates and large standard errors even when the fitted values are more stable. High-leverage observations, nonlinear relationships, and the choice of important predictors also require attention. Variable selection asks which predictors should be included; training fit alone cannot answer this question.

Chapter 4

Generative Classification and Discriminant Analysis

4.1 Prior probabilities, class-conditional distributions, and posteriors

Let $Y \in \{1, \dots, K\}$ and $X \in \mathbb{R}^p$. Recall that the Bayes classifier chooses the class with largest posterior probability

$$P_k(x) = \mathbb{P}(Y = k \mid X = x), \qquad \sum_{k=1}^{K} P_k(x) = 1.$$

The prior probability of class k is

$$\pi_k = \mathbb{P}(Y = k), \qquad \sum_{k=1}^{K} \pi_k = 1.$$

A prior is marginal; a posterior is conditional on the predictors.

The class-conditional distribution is the distribution of X for an observation from class k. For discrete predictors, write $f_k(x) = \mathbb{P}(X = x \mid Y = k)$; for continuous predictors, let f_k be the conditional joint density of $X \mid Y = k$. Assume that the class distributions have densities or mass functions with respect to a common measure. Bayes' theorem gives

$$P_k(x) = \frac{\pi_k f_k(x)}{\sum_{\ell=1}^{K} \pi_\ell f_\ell(x)}$$

whenever the denominator is positive. For discrete X, the calculation is

$$\begin{aligned} \mathbb{P}(Y = k \mid X = x) &= \frac{\mathbb{P}(X = x, Y = k)}{\mathbb{P}(X = x)} \\ &= \frac{\mathbb{P}(X = x \mid Y = k)\mathbb{P}(Y = k)}{\sum_{\ell=1}^{K} \mathbb{P}(X = x \mid Y = \ell)\mathbb{P}(Y = \ell)}. \end{aligned}$$

The same calculation uses densities for continuous X.

The denominator is a normalizing factor common to all classes. Thus class comparisons need only compare $\pi_k f_k(x)$. A *generative* method models the class-conditional distributions f_k and uses

Bayes' theorem to obtain posterior probabilities. Discriminant analysis and naive Bayes follow this approach. Logistic regression models posterior probabilities directly and can be used for both prediction and inference.

Estimating an unrestricted joint distribution in many dimensions is difficult. Two simplifying approaches are to assume multivariate normal class conditional distributions, or to assume that predictor components are conditionally independent within each class.

4.2 Random vectors and the multivariate normal distribution

For a random vector $X = (X_1, \dots, X_p)^T$ with finite second moments, its mean vector is $\mu = \mathbb{E}(X)$ and its covariance matrix is $\Sigma = \mathrm{Var}(X)$.

Exercise 4.1. Write the entries of μ and Σ, and calculate the mean and covariance of linear combinations $U = a^T X$ and $V = b^T X$.

Solution. The entries are

$$\mu_j = \mathbb{E}(X_j), \qquad \Sigma_{jj} = \mathrm{Var}(X_j), \qquad \Sigma_{ij} = \mathrm{Cov}(X_i, X_j).$$

Equivalently,

$$\Sigma = \mathbb{E}\{(X - \mu)(X - \mu)^T\}.$$

It is symmetric because $\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$. A diagonal covariance matrix means that distinct components are uncorrelated. By linearity,

$$\mathbb{E}(U) = a^T \mu.$$

Furthermore,

$$\begin{aligned} \mathrm{Cov}(U, V) &= \mathbb{E}\{a^T (X - \mu)(X - \mu)^T b\} \\ &= a^T \Sigma b. \end{aligned}$$

Taking $b = a$ gives $\mathrm{Var}(U) = a^T \Sigma a$.

The covariance matrix is symmetric and positive semidefinite, because for every $a \in \mathbb{R}^p$,

$$a^T \Sigma a = \mathrm{Var}(a^T X) \geq 0.$$

In particular, every eigenvalue of Σ is nonnegative: if $\Sigma v = \rho v$ with $v \neq 0$, then $\rho \|v\|^2 = v^T \Sigma v \geq 0$.

For $X \sim N_p(\mu, \Sigma)$ with positive definite Σ, the density is

$$f(x) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left\{ -\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right\}.$$

Every linear combination is normal:

$$a^T X \sim N(a^T \mu, a^T \Sigma a).$$

For a jointly multivariate normal vector, uncorrelated components are independent. This implication need not hold for other joint distributions. The covariance matrix is also called the variance–covariance matrix; its diagonal entries are variances and its off-diagonal entries are covariances.

4.3 Discriminant functions

For two classes, the Bayes rule assigns x to class 1 if

$$P_1(x) > P_2(x) \quad \Longleftrightarrow \quad \pi_1 f_1(x) > \pi_2 f_2(x).$$

If the priors and densities are positive, the logarithm is strictly increasing, so we may compare the *discriminant functions*

$$\delta_k(x) = \log\{\pi_k f_k(x)\}.$$

The posterior probabilities and discriminant functions induce the same ordering of classes. Assign class 1 when $\delta_1(x) > \delta_2(x)$ and class 2 when the inequality is reversed, with a fixed rule for ties. The decision boundary is $\{x : \delta_1(x) = \delta_2(x)\}$.

Suppose now that

$$X \mid Y = k \sim N_p(\mu_k, \Sigma_k), \qquad \Sigma_k \text{ positive definite.}$$

Then

$$\begin{aligned}\log\{\pi_k f_k(x)\} = \log \pi_k - \frac{p}{2}\log(2\pi) - \frac{1}{2}\log|\Sigma_k| \\ - \frac{1}{2}(x-\mu_k)^T\Sigma_k^{-1}(x-\mu_k).\end{aligned}$$

The quadratic form expands as

$$(x-\mu_k)^T\Sigma_k^{-1}(x-\mu_k) = x^T\Sigma_k^{-1}x - 2\mu_k^T\Sigma_k^{-1}x + \mu_k^T\Sigma_k^{-1}\mu_k.$$

Class-specific covariance matrices generally give quadratic discriminant functions. With a common covariance matrix, the quadratic term is identical for all classes and disappears from class comparisons.

4.4 Linear discriminant analysis

Linear discriminant analysis (LDA) assumes

$$X \mid Y = k \sim N_p(\mu_k, \Sigma), \qquad k = 1, \dots, K.$$

Removing all terms common to the classes gives the equivalent scores

$$\delta_k^{\mathrm{L}}(x) = x^T\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k + \log \pi_k.$$

These are linear functions of the predictors, including a constant term.

4.4.1 Estimating the model

Given training pairs (X_i, Y_i), define

$$n_k = \sum_{i=1}^{n} \mathbf{1}\{Y_i = k\}, \qquad \bar{X}_k = \frac{1}{n_k}\sum_{i:Y_i=k} X_i.$$

For $n_k > 1$, the within-class sample covariance is

$$S_k = \frac{1}{n_k - 1} \sum_{i:Y_i=k} (X_i - \bar{X}_k)(X_i - \bar{X}_k)^T.$$

Natural estimates are

$$\hat{\pi}_k = \frac{n_k}{n}, \qquad \hat{\mu}_k = \bar{X}_k, \qquad \hat{\Sigma} = \frac{\sum_{k=1}^{K}(n_k - 1)S_k}{n - K}.$$

The covariance estimate pools the within-class covariances. For two classes,

$$\hat{\Sigma} = \frac{(n_1 - 1)S_1 + (n_2 - 1)S_2}{n - 2}.$$

Use these estimates in the discriminant functions and assign the class with largest estimated score. The usual inverse formulas require the pooled covariance matrix to be nonsingular.

Exercise 4.2. For two classes, express the LDA rule as a linear inequality $\hat{a}^T x > \hat{c}$. Simplify it when the estimated priors are equal.

Solution. Subtract the estimated scores:

$$\begin{aligned}\hat{\delta}_1^{\mathrm{L}}(x) - \hat{\delta}_2^{\mathrm{L}}(x) &= x^T \hat{\Sigma}^{-1}(\hat{\mu}_1 - \hat{\mu}_2) \\ &\quad - \frac{1}{2}\left(\hat{\mu}_1^T \hat{\Sigma}^{-1}\hat{\mu}_1 - \hat{\mu}_2^T \hat{\Sigma}^{-1}\hat{\mu}_2\right) + \log\frac{\hat{\pi}_1}{\hat{\pi}_2}.\end{aligned}$$

Thus set

$$\begin{aligned}\hat{a} &= \hat{\Sigma}^{-1}(\hat{\mu}_1 - \hat{\mu}_2), \\ \hat{c} &= \frac{1}{2}\left(\hat{\mu}_1^T \hat{\Sigma}^{-1}\hat{\mu}_1 - \hat{\mu}_2^T \hat{\Sigma}^{-1}\hat{\mu}_2\right) - \log\frac{\hat{\pi}_1}{\hat{\pi}_2}.\end{aligned}$$

Class 1 is chosen when $\hat{a}^T x > \hat{c}$; the boundary is the hyperplane $\{x : \hat{a}^T x = \hat{c}\}$ when $\hat{a} \neq 0$. For equal estimated priors, symmetry of $\hat{\Sigma}^{-1}$ gives

$$\hat{c} = \frac{1}{2}(\hat{\mu}_1 + \hat{\mu}_2)^T \hat{\Sigma}^{-1}(\hat{\mu}_1 - \hat{\mu}_2).$$

Equivalently, choose the mean closest to x in squared Mahalanobis distance:

$$(x - \hat{\mu}_1)^T \hat{\Sigma}^{-1}(x - \hat{\mu}_1) < (x - \hat{\mu}_2)^T \hat{\Sigma}^{-1}(x - \hat{\mu}_2).$$

If the means coincide, the prior probabilities alone determine the decision, with ties when the priors are also equal.

LDA is an estimated Bayes classifier under multivariate normality and common within-class covariance. The linear boundary results from the common covariance assumption, not from assuming that the marginal distribution of all predictors is a single normal distribution.

4.5 Quadratic discriminant analysis

Quadratic discriminant analysis (QDA) allows a different covariance matrix for each class:

$$X \mid Y = k \sim N_p(\mu_k, \Sigma_k).$$

After removing only the common term $-(p/2)\log(2\pi)$, use

$$\delta_k^{\mathrm{Q}}(x) = -\frac{1}{2}\log|\Sigma_k| - \frac{1}{2}(x-\mu_k)^T\Sigma_k^{-1}(x-\mu_k) + \log\pi_k.$$

Estimate π_k and μ_k as before and estimate Σ_k by S_k. The plug-in classifier chooses the largest estimated score. These formulas require nonsingular class covariance estimates, which need enough independent within-class variation in every predictor direction.

Exercise 4.3. Write the two-class QDA rule explicitly as a quadratic inequality and identify its decision boundary.

Solution. For simplicity write the population parameters; the fitted rule replaces each one by its estimate. Subtracting the two scores gives

$$\begin{aligned}\delta_1^{\mathrm{Q}}(x) - \delta_2^{\mathrm{Q}}(x) = {} & \frac{1}{2}x^T(\Sigma_2^{-1} - \Sigma_1^{-1})x \\ & + (\Sigma_1^{-1}\mu_1 - \Sigma_2^{-1}\mu_2)^T x \\ & - \frac{1}{2}\left(\mu_1^T\Sigma_1^{-1}\mu_1 - \mu_2^T\Sigma_2^{-1}\mu_2\right) \\ & - \frac{1}{2}\log\frac{|\Sigma_1|}{|\Sigma_2|} + \log\frac{\pi_1}{\pi_2}.\end{aligned}$$

Assign class 1 when this expression is positive, and class 2 when it is negative. Its zero set is the decision boundary, generally a quadratic surface. If $\Sigma_1 = \Sigma_2$, the quadratic and log-determinant terms cancel, recovering the LDA rule.

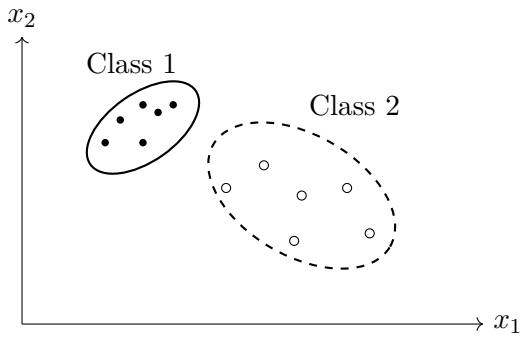
4.5.1 More than two classes and the bias–variance trade-off

For $K > 2$, assign x to the class whose posterior probability, or equivalently whose discriminant score, is largest. The region for class k is

$$R_k = \{x : \delta_k(x) > \delta_\ell(x) \text{ for every } \ell \neq k\},$$

with ties assigned by a fixed convention. Pairwise equal-score surfaces form the possible boundaries; a pairwise equality is an active boundary only where neither class is dominated by a third class. Estimated scores give estimated regions.

LDA and QDA are both generative normal models. Their central distinction is whether covariance is common across classes. Different shapes, spreads, or orientations of class clusters suggest the extra flexibility of QDA:



Class clusters with different covariance structures.

QDA can also be used when the true covariances coincide, but the simpler LDA model may estimate more reliably. LDA is less flexible, typically with lower variance and higher bias when its covariance restriction is false. QDA estimates more parameters and is more sensitive to limited class sample sizes and departures from its normal modeling assumptions. The choice reflects a bias–variance trade-off rather than a rule that the more flexible model must predict better.

4.5.2 Parameter counts

A symmetric $p \times p$ covariance matrix has

$$p + \frac{p(p-1)}{2} = \frac{p(p+1)}{2}$$

free entries. Thus, with K classes, the parameter counts are

	LDA	QDA
Means	Kp	Kp
Covariances	$p(p+1)/2$	$Kp(p+1)/2$
Priors	$K-1$	$K-1$

when the priors are estimated. LDA pools within-class covariance information; QDA estimates a covariance matrix separately in each class. The extra covariance parameters explain why QDA needs more data for stable estimation. For the unbiased pooled covariance, the residual degrees of freedom are $n - K$, equal to $n - 2$ in the binary case.

4.6 Naive Bayes

Naive Bayes assumes that, conditional on $Y = k$, the p components of X are independent. It factors the class-conditional joint density or mass function as

$$f_k(x_1, \ldots, x_p) = \prod_{j=1}^{p} f_{kj}(x_j).$$

Here f_{kj} is the distribution of predictor j within class k. This turns the difficult joint estimation problem into several one-dimensional problems. The independence is conditional on the class; it is not an assumption of unconditional independence of the predictors.

Estimate the priors by $\hat{\pi}_k = n_k/n$. A parametric choice for a continuous predictor is

$$X_j \mid Y = k \sim N(\mu_{jk}, \sigma_{jk}^2),$$

with its mean and variance estimated from the observations in class k. A nonparametric alternative is a class-specific histogram for each predictor. For discrete predictors, estimate the corresponding class-specific mass functions. The class comparison uses

$$\hat{\pi}_k \prod_{j=1}^{p} \hat{f}_{kj}(x_j),$$

or its logarithm. Normalizing these quantities over classes produces estimated posterior probabilities.

4.6.1 Histogram estimates

Let a histogram bin B have width $w_B > 0$. If n_k observations belong to class k, its estimated probability is

$$\widehat{\mathbb{P}}(X_j \in B \mid Y = k) = \frac{\#\{i : Y_i = k,\ X_{ij} \in B\}}{n_k}.$$

For a continuous predictor the histogram density equals this fraction divided by w_B. For a discrete predictor, the empirical fraction at a value estimates a probability mass directly. These one-dimensional estimates are combined with the estimated priors and normalized over the classes to obtain posterior probabilities.

4.7 Measuring classification performance

Overall misclassification probability can conceal very different class-specific errors. In a two-class problem,

$$\begin{aligned}\mathbb{P}(\hat{Y} \neq Y) &= \mathbb{P}(\hat{Y} \neq Y, Y = 1) + \mathbb{P}(\hat{Y} \neq Y, Y = 2) \\ &= \pi_1 \mathbb{P}(\hat{Y} = 2 \mid Y = 1) + \pi_2 \mathbb{P}(\hat{Y} = 1 \mid Y = 2).\end{aligned}$$

The overall error combines the two conditional error rates using the class priors.

Example 4.1 (An imbalanced population)**.** Suppose a hypothetical condition is present in 2% of a population. Among 100 observations with exactly two positive cases, a classifier that always predicts negative makes only two errors, for an overall error rate of 2%. However,

$$\mathbb{P}(\hat{Y} = + \mid Y = -) = 0, \qquad \mathbb{P}(\hat{Y} = - \mid Y = +) = 1.$$

It misses every positive case. Its small overall error is therefore an incomplete description of its performance. Class-specific rates must also be examined, particularly when one class dominates.

Treating a classifier as a diagnostic test, call the positive class the non-null class and the negative class the null class. Define

$$\begin{aligned}\text{sensitivity} &= \mathbb{P}(\hat{Y} = + \mid Y = +), \\ \text{specificity} &= \mathbb{P}(\hat{Y} = - \mid Y = -).\end{aligned}$$

These are the true-positive and true-negative rates. The false-positive rate is $1 - \text{specificity}$ and the false-negative rate is $1 - \text{sensitivity}$. If $\pi_+ = \mathbb{P}(Y = +)$ and $\pi_- = \mathbb{P}(Y = -)$, then

$$\mathbb{P}(\hat{Y} \neq Y) = (1 - \text{sensitivity})\pi_+ + (1 - \text{specificity})\pi_-.$$

Both sensitivity and specificity should ideally be high. With imbalanced classes, minimizing unweighted zero–one loss can favor the dominant class; whether that is appropriate depends on the purpose and the costs of the two types of error.

4.7.1 Confusion matrices

For a binary classifier, distinguish observed counts from conditional rates. A confusion matrix can be written as

	$\hat{Y} = -$	$\hat{Y} = +$	Total
$Y = -$	TN	FP	N
$Y = +$	FN	TP	P
Total	N^*	P^*	$N + P$

Here P and N count the actual positive and negative observations, and P^* and N^* count predicted positives and negatives. Provided both actual classes occur,

$$\widehat{\text{sensitivity}} = \frac{\text{TP}}{P}, \qquad \widehat{\text{specificity}} = \frac{\text{TN}}{N},$$
$$\widehat{\text{FNR}} = \frac{\text{FN}}{P}, \qquad \widehat{\text{FPR}} = \frac{\text{FP}}{N}.$$

The two rates conditional on actual positives sum to one, and so do the two rates conditional on actual negatives:

$$\frac{\text{TP}}{P} + \frac{\text{FN}}{P} = 1, \qquad \frac{\text{TN}}{N} + \frac{\text{FP}}{N} = 1.$$

The empirical misclassification rate is

$$\begin{aligned}\widehat{\mathbb{P}}(\hat{Y} \neq Y) &= \frac{\text{FN}}{P}\frac{P}{N+P} + \frac{\text{FP}}{N}\frac{N}{N+P} \\ &= \frac{\text{FN} + \text{FP}}{N+P}.\end{aligned}$$

Thus a low overall error may hide a high error rate within a rare class.

4.8 ROC curves and area under the curve

Let $s(x)$ be a score, with larger values indicating stronger evidence for the positive class. A posterior probability estimate is one such score. Predict positive if $s(x) \geq t$. As t increases, the positive prediction set shrinks, so sensitivity cannot increase and specificity cannot decrease. These changes can include flat portions; they need not be strict.

If $0 < s(x) < 1$, thresholds $t = 0$ and $t = 1$ give the always-positive and always-negative rules. When scores can equal an endpoint, the tie rule matters: under $s(x) \geq t$, a score of exactly one is positive even at $t = 1$. An always-negative rule can instead use a threshold above the largest score.

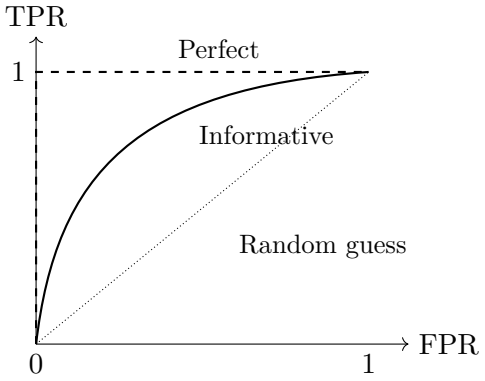
For fitted probabilities strictly between zero and one, the endpoint rules are

Threshold	Specificity	Sensitivity
0	0	1
1	1	0

The receiver operating characteristic curve plots

$$(\mathrm{FPR}(t), \mathrm{TPR}(t)) = (1 - \mathrm{specificity}(t), \mathrm{sensitivity}(t))$$

as the threshold varies. It can be estimated using a fine grid of thresholds, or the distinct observed scores. The following diagrams show the ideal curve, an informative classifier, and random guessing.



A perfect classifier has the inverted L-shaped curve and area one. A random guess made independently of the data, positive with probability q, has sensitivity q and false-positive rate q. Varying q traces the diagonal, whose area is $1/2$.

The area under the ROC curve, or AUC, summarizes discrimination over thresholds. If S_+ and S_- are independent scores from a positive and a negative observation, respectively, the usual convention is

$$\mathrm{AUC} = \mathbb{P}(S_+ > S_-) + \tfrac{1}{2}\mathbb{P}(S_+ = S_-).$$

Thus AUC assesses the ordering of the scores, rather than the correctness of two hard class labels at one cutoff.

Exercise 4.4. Suppose the positive and negative score distributions are continuous. Explain why ROC area equals $\mathbb{P}(S_+ > S_-)$, and give its empirical form when ties are allowed.

Solution. Let F_+ and F_- be the score distribution functions. At threshold t, continuity gives

$$\mathrm{TPR}(t) = 1 - F_+(t), \qquad \mathrm{FPR}(t) = 1 - F_-(t).$$

As the threshold moves from high to low, the area under this parameterized curve is

$$\mathrm{AUC} = \int_{-\infty}^{\infty} [1 - F_+(t)]\, dF_-(t).$$

Conditioning on the independent negative score yields

$$\mathbb{P}(S_+ > S_-) = \mathbb{E}\{\mathbb{P}(S_+ > S_- \mid S_-)\} = \int [1 - F_+(t)]\, dF_-(t),$$

which is the same quantity. With P positive scores s_i^+ and N negative scores s_j^-, the empirical version, giving half credit to ties, is

$$\widehat{\mathrm{AUC}} = \frac{1}{PN} \sum_{i=1}^{P} \sum_{j=1}^{N} \left[\mathbf{1}\{s_i^+ > s_j^-\} + \frac{1}{2} \mathbf{1}\{s_i^+ = s_j^-\} \right].$$

The half-credit convention agrees with linearly interpolating the empirical ROC at tied scores.

ROC curves and AUC can be used to compare classifiers. If one ROC curve lies entirely above another, it gives at least as much sensitivity at the same false-positive rate throughout the comparison. Curves can also cross, in which case the preferred classifier can depend on the relevant operating range. In practice, inspect the curve and use knowledge of the application to choose an acceptable sensitivity and specificity. AUC by itself does not select a cutoff.

Chapter 5

Logistic Regression and Direct Probability Models

5.1 Link functions for a binary response

Discriminant analysis and naive Bayes model $f_k(x)$ and use Bayes' theorem to obtain $P_k(x)$. Logistic regression models the posterior class probability directly. Code the two classes as 0 and 1 and write

$$Y \mid X = x \sim \text{Bernoulli}\{p(x)\}, \qquad \mathbb{E}(Y \mid X = x) = p(x).$$

A modeling principle is to find an increasing function g of the conditional mean with real-valued range and set

$$g\{\mathbb{E}(Y \mid X = x)\} = x^T\beta,$$

where $x = (1, x_1, \ldots, x_p)^T$. The ordinary linear model uses the identity link $g(u) = u$. For a probability, the direct model $p(x) = x^T\beta$ has no guarantee of remaining between zero and one for all x.

The *probit* link uses the standard normal distribution function Φ:

$$\Phi^{-1}\{p(x)\} = x^T\beta, \qquad p(x) = \Phi(x^T\beta) \in (0, 1).$$

The *logit* link is

$$\text{logit}(p) = \log \frac{p}{1-p}, \qquad 0 < p < 1.$$

It maps $(0, 1)$ onto $\mathbb{R}$. Logistic regression assumes

$$\log \frac{p(x)}{1-p(x)} = x^T\beta.$$

Solving for $p(x)$ gives the logistic function

$$\frac{p(x)}{1-p(x)} = \exp(x^T\beta), \qquad p(x) = \frac{\exp(x^T\beta)}{1+\exp(x^T\beta)} \in (0, 1).$$

The linear structure is on the log-odds scale, rather than on the probability or conditional-mean scale. Predictors can be continuous or discrete.

5.2 Interpreting coefficients through odds

The odds corresponding to probability p are $p/(1-p)$. They approach zero as $p \to 0$ and infinity as $p \to 1$. With one predictor,

$$\log \frac{p(x)}{1-p(x)} = \beta_0 + \beta_1 x.$$

A one-unit increase in x increases the log odds by β_1. Subtracting the log odds at x from those at $x+1$ gives

$$\log \left\{ \frac{p(x+1)/(1-p(x+1))}{p(x)/(1-p(x))} \right\} = \beta_1.$$

Therefore

$$\frac{p(x+1)/(1-p(x+1))}{p(x)/(1-p(x))} = \exp(\beta_1),$$

so the new odds equal the old odds multiplied by e^{β_1}. If $\beta_1 > 0$, both odds and probability increase with x; if $\beta_1 < 0$, both decrease. If $\beta_1 = 0$, the probability does not depend on x in this one-predictor model. With several predictors, the interpretation holds while the remaining predictor values are fixed.

An odds ratio is not generally a probability ratio. If both probabilities are small, $1-p(x)$ and $1-p(x+1)$ are near one, and

$$\frac{p(x+1)/(1-p(x+1))}{p(x)/(1-p(x))} \approx \frac{p(x+1)}{p(x)}.$$

Only in this rare-outcome regime is the odds ratio approximately a relative risk.

Example 5.1 (Smoking categories)**.** Suppose a logistic model uses never-smokers as the reference category and has indicator coefficients $\beta_1 = 1.1$ for occasional smoking and $\beta_2 = 3.0$ for serious smoking. Holding any other predictors fixed,

$$e^{1.1} \approx 3.0, \qquad e^{3.0} \approx 20.1, \qquad e^{3.0-1.1} = e^{1.9} \approx 6.7.$$

The modeled outcome odds for an occasional smoker are about three times those for a never-smoker. For a serious smoker they are about twenty times those for a never-smoker, and about 6.7 times those for an occasional smoker. These statements concern odds; they do not specify the outcome's baseline probability.

5.3 Fitting and the decision boundary

Estimate β from the training data and plug it into the probability model:

$$\text{logit}\{\hat{p}(x)\} = x^T \hat{\beta}, \qquad \hat{p}(x) = \frac{\exp(x^T \hat{\beta})}{1 + \exp(x^T \hat{\beta})}.$$

Under equal misclassification costs, the plug-in classification rule predicts class 1 when $\hat{p}(x) > 1/2$ and class 0 otherwise, subject to a fixed tie rule. Because the logit is increasing and $\text{logit}(1/2) = 0$,

$$\hat{p}(x) > 1/2 \quad \Longleftrightarrow \quad x^T \hat{\beta} > 0.$$

The decision boundary is

$$\{x : \hat{p}(x) = 1/2\} = \{x : x^T \hat{\beta} = 0\}.$$

It is a hyperplane in the predictor coordinates entering the model. An estimated probability model gives a plug-in Bayes rule; it is the true Bayes rule only when its probabilities induce the true posterior ordering.

5.3.1 A probability calculation

Exercise 5.1. Suppose a fitted logistic regression model is

$$\log \frac{p(x_1, x_2)}{1 - p(x_1, x_2)} = -6 + 0.05x_1 + 1.5x_2.$$

For $x_2 = 3.5$, express the fitted probability as a function of x_1 and find the value of x_1 at which it equals $1/2$.

Solution. Substitution gives

$$\eta = -6 + 0.05x_1 + 1.5(3.5) = -0.75 + 0.05x_1.$$

Since $p/(1-p) = e^{\eta}$, solving for p gives

$$p(x_1, 3.5) = \frac{e^{-0.75+0.05x_1}}{1 + e^{-0.75+0.05x_1}} = \frac{1}{1 + e^{0.75-0.05x_1}}.$$

The probability is $1/2$ precisely when its log odds are zero. Thus

$$-0.75 + 0.05x_1 = 0, \qquad x_1 = \frac{0.75}{0.05} = 15.$$

5.3.2 Discrete response distributions

For a single binary outcome,

$$Y \sim \text{Bernoulli}(p), \qquad \mathbb{P}(Y = y) = p^y(1-p)^{1-y}, \qquad y = 0, 1.$$

For the sum of m independent Bernoulli outcomes sharing the same probability,

$$Y \sim \text{Binomial}(m, p), \qquad \mathbb{P}(Y = y) = \binom{m}{y} p^y (1-p)^{m-y}, \qquad y = 0, \ldots, m.$$

Exercise 5.2. Write the likelihood for independent binary observations under logistic regression. Extend the Bernoulli and binomial mass functions to K categories.

Solution. Let $p_i = p(x_i)$. Conditional independence gives

$$L(\beta) = \prod_{i=1}^{n} p_i^{y_i} (1 - p_i)^{1-y_i}.$$

Substituting $p_i = e^{x_i^T\beta}/(1+e^{x_i^T\beta})$ gives

$$\ell(\beta) = \sum_{i=1}^{n} \left[y_i x_i^T \beta - \log\{1 + \exp(x_i^T \beta)\} \right].$$

A finite maximizer, when it exists, is the maximum likelihood coefficient estimate.

For a single categorical outcome, let $Z_k = \mathbf{1}\{Y = k\}$ with exactly one component equal to one. If the class probabilities are $p_1, \ldots, p_K$ and sum to one, its categorical, or multinoulli, mass function is

$$\mathbb{P}(Z = z) = \prod_{k=1}^{K} p_k^{z_k}, \qquad z_k \in \{0, 1\}, \quad \sum_k z_k = 1.$$

For m independent trials with those probabilities, the category counts $N_1, \ldots, N_K$ have multinomial mass function

$$\mathbb{P}(N_1 = n_1, \ldots, N_K = n_K) = \frac{m!}{\prod_k n_k!} \prod_{k=1}^{K} p_k^{n_k}, \qquad n_k \geq 0, \quad \sum_k n_k = m.$$

For $K = 2$, the first component has the binomial distribution, and $m = 1$ recovers the categorical distribution.

5.4 Multinomial logistic regression and softmax

For $K > 2$, choose class K as the baseline and model the $K - 1$ log odds relative to it.

Exercise 5.3. If

$$\log \frac{\mathbb{P}(Y = k \mid X = x)}{\mathbb{P}(Y = K \mid X = x)} = x^T \beta_k, \qquad k = 1, \ldots, K-1,$$

recover all class probabilities and the classification rule.

Solution. Write $p_k(x) = \mathbb{P}(Y = k \mid X = x)$. Exponentiating gives $p_k(x) = e^{x^T\beta_k} p_K(x)$ for $k < K$. Since the probabilities sum to one,

$$p_K(x) \left(1 + \sum_{\ell=1}^{K-1} e^{x^T \beta_\ell} \right) = 1.$$

Thus

$$p_K(x) = \frac{1}{1 + \sum_{\ell=1}^{K-1} e^{x^T\beta_\ell}},$$
$$p_k(x) = \frac{e^{x^T\beta_k}}{1 + \sum_{\ell=1}^{K-1} e^{x^T\beta_\ell}}, \qquad k < K.$$

Setting $\beta_K = 0$ expresses these in softmax form:

$$p_k(x) = \frac{\exp(x^T\beta_k)}{\sum_{\ell=1}^{K} \exp(x^T\beta_\ell)}, \qquad k = 1, \ldots, K.$$

Fixing the baseline removes the nonidentifiability caused by adding the same coefficient vector to every β_k. Estimate the coefficients and select the largest fitted probability, equivalently the largest fitted linear score $x^T\hat{\beta}_k$. Pairwise equal-score boundaries satisfy $x^T(\hat{\beta}_k - \hat{\beta}_\ell) = 0$.

5.5 Analytical comparison with discriminant analysis

Logistic regression does not prescribe a distribution for the predictors. Its assumption concerns the conditional class probabilities. LDA and QDA instead prescribe normal distributions within each class; naive Bayes prescribes conditional independence and models the individual predictor distributions.

Exercise 5.4. Show that two-class LDA implies a logistic posterior probability. Identify the intercept and slope coefficients, and compare with QDA.

Solution. Use labels 1 and 2, and let $p(x) = \mathbb{P}(Y = 1 \mid X = x)$. Bayes' theorem gives

$$\log \frac{p(x)}{1 - p(x)} = \log \frac{\pi_1 f_1(x)}{\pi_2 f_2(x)}.$$

With normal class distributions and a common covariance,

$$\begin{aligned} \log \frac{p(x)}{1 - p(x)} &= \log \frac{\pi_1}{\pi_2} - \frac{1}{2} \left\{ (x - \mu_1)^T \Sigma^{-1} (x - \mu_1) - (x - \mu_2)^T \Sigma^{-1} (x - \mu_2) \right\} \\ &= a + b^T x, \end{aligned}$$

where, here using x for the p predictors without an intercept coordinate,

$$b = \Sigma^{-1}(\mu_1 - \mu_2), \qquad a = \log \frac{\pi_1}{\pi_2} - \frac{1}{2} \left(\mu_1^T \Sigma^{-1} \mu_1 - \mu_2^T \Sigma^{-1} \mu_2 \right).$$

Consequently,

$$p(x) = \frac{e^{a + b^T x}}{1 + e^{a + b^T x}}.$$

Thus LDA's population posterior has a linear logistic form. LDA and logistic regression can nevertheless produce different fitted coefficients because they estimate parameters using different assumptions and likelihoods. With distinct covariance matrices, the log odds equal the QDA score difference and are generally quadratic in x. They can be represented by a logistic model that includes the required squared and product terms.

Logistic regression can be unstable with small samples. If the classes are perfectly separated by a linear score, ordinary maximum likelihood can fail to have a finite coefficient estimate: scaling a separating coefficient vector can continually increase the likelihood. Good apparent separation therefore does not by itself establish stable probability estimates.

Chapter 6

Generalized Linear Models and Likelihood Inference

6.1 The three components of a model

The normal linear model has three components. Conditional on the predictors,

$$Y_i \sim N(\mu_i, \sigma^2), \qquad \eta_i = x_i^T \beta, \qquad \eta_i = g(\mu_i) = \mu_i.$$

The distribution is the random component, η_i is the systematic or linear predictor, and the identity function g links the mean to that predictor. Here and below the design vector x_i includes a leading one when an intercept is fitted. Responses are assumed conditionally independent.

A generalized linear model extends the random component to an exponential family distribution and permits a different link. In the increasing-link convention, g is differentiable and increasing on the mean's parameter space, and

$$g(\mu_i) = x_i^T \beta, \qquad \mu_i = \mathbb{E}(Y_i \mid x_i).$$

The systematic component remains linear in β. The distribution and link must be chosen so that $g^{-1}(x_i^T \beta)$ is a valid mean.

For example, a binary response satisfies $Y_i \mid x_i \sim \text{Bernoulli}(p_i)$ and $\mu_i = p_i$. The identity link would set $p_i = x_i^T \beta$, which need not stay in $[0, 1]$. The probit link is $g(p) = \Phi^{-1}(p)$, and the logit link is

$$g(p) = \log \frac{p}{1-p}, \qquad p_i = \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)}.$$

For a one-predictor logistic model, increasing x by one multiplies the odds by e^{β_1}. These links model the conditional probability directly; discriminant analysis instead obtains it from class-conditional predictor distributions and the priors.

Other common links include the log link $g(\mu) = \log \mu$, appropriate for a positive mean, and the complementary log–log link

$$g(p) = \log\{-\log(1-p)\}, \qquad 0 < p < 1.$$

A Poisson response with a log link is another example of a GLM.

6.2 Exponential-family form

A density or probability mass function is in exponential-family form if it can be expressed as

$$f_Y(y;\theta,\phi) = \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right\}.$$

The parameter θ is the canonical or natural parameter, ϕ is a dispersion parameter, and the functions a, b, c specify the family. In a GLM, the observations can have different θ_i through their predictors. A canonical link equates θ_i to the linear predictor η_i.

Exercise 6.1. Express a normal density with mean μ and variance $\sigma^2 > 0$ in this form, identifying all its terms.

Solution. Expand the square in the normal density:

$$\begin{aligned} f_Y(y) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(y-\mu)^2}{2\sigma^2}\right\} \\ &= \exp\left\{-\frac{y^2 - 2y\mu + \mu^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)\right\} \\ &= \exp\left\{\frac{y\mu - \mu^2/2}{\sigma^2} - \frac{y^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)\right\}. \end{aligned}$$

Thus we can take

$$\theta = \mu, \quad b(\theta) = \frac{\theta^2}{2}, \quad \phi = \sigma^2, \quad a(\phi) = \phi, \quad c(y,\phi) = -\frac{y^2}{2\phi} - \frac{1}{2}\log(2\pi\phi).$$

The canonical link is the identity link.

Example 6.1 (Bernoulli exponential family)**.** For $y \in \{0, 1\}$ and $0 < p < 1$,

$$\begin{aligned} f_Y(y) &= p^y (1-p)^{1-y} \\ &= \exp\{y \log p + (1-y)\log(1-p)\} \\ &= \exp\left\{y \log\frac{p}{1-p} + \log(1-p)\right\}. \end{aligned}$$

Set $\theta = \log\{p/(1-p)\}$. Then $p = e^\theta/(1+e^\theta)$ and $\log(1-p) = -\log(1+e^\theta)$, so

$$f_Y(y) = \exp\{y\theta - \log(1+e^\theta)\}.$$

Here $b(\theta) = \log(1+e^\theta)$, $a(\phi) = 1$, and $c(y,\phi) = 0$. The dispersion is fixed at one, and the logit is the canonical link.

6.3 Maximum likelihood and coefficient uncertainty

A binary response has conditional variance $p_i(1-p_i)$, rather than a constant normal error variance. Thus the usual normal least-squares likelihood and its exact t intervals do not apply to logistic

regression. The appropriate likelihood uses the chosen response distribution. Let $\ell(\beta, \phi)$ be the conditional log likelihood. Its maximizer gives $\hat{\beta}$, and the fitted means are

$$\hat{\mu}_i = g^{-1}(x_i^T \hat{\beta}), \qquad i = 1, \ldots, n.$$

Under regularity conditions, with an identifiable fixed-dimensional parameter, a finite interior maximum, and a sample large enough for the approximation,

$$\hat{\beta} \;\dot{\sim}\; N(\beta, \widehat{\mathcal{J}}^{-1}), \qquad \widehat{\mathcal{J}} = - \left. \frac{\partial^2 \ell}{\partial \beta \, \partial \beta^T} \right|_{(\hat{\beta}, \hat{\phi})}.$$

Here $\widehat{\mathcal{J}}$ is the observed information, the negative Hessian of the log likelihood with respect to the coefficients. For models with an estimated nuisance dispersion, use the appropriate coefficient covariance block of the full information inverse; the displayed coefficient formula applies when that nuisance estimation is orthogonal to the mean parameters or the dispersion is known. A Wald standard error is the square root of a diagonal entry of the estimated coefficient covariance.

Exercise 6.2. For independent Bernoulli responses with the logit link, derive the score, observed information, and a large-sample confidence interval for a coefficient.

Solution. Write $\eta_i = x_i^T \beta$ and $p_i = e^{\eta_i}/(1 + e^{\eta_i})$. The log likelihood is

$$\ell(\beta) = \sum_{i=1}^{n} \{ y_i \eta_i - \log(1 + e^{\eta_i}) \}.$$

Since $\partial p_i / \partial \eta_i = p_i(1 - p_i)$,

$$\frac{\partial \ell}{\partial \beta} = \sum_i x_i (y_i - p_i) = X^T (Y - p),$$
$$\frac{\partial^2 \ell}{\partial \beta \, \partial \beta^T} = - \sum_i p_i (1 - p_i) x_i x_i^T = -X^T W X,$$

where $W = \operatorname{diag}\{p_i(1 - p_i)\}$. A finite maximum solves $X^T(Y - \hat{p}) = 0$; numerical iteration is generally required. At the maximum,

$$\widehat{\mathcal{J}} = X^T \widehat{W} X, \qquad \widehat{\operatorname{Var}}(\hat{\beta}) = (X^T \widehat{W} X)^{-1}.$$

Let v_j be the diagonal entry of this inverse corresponding to $\hat{\beta}_j$ and let $z_{1-\alpha/2}$ be a standard normal quantile. An approximate $1 - \alpha$ interval is

$$\hat{\beta}_j \;\pm\; z_{1-\alpha/2} \sqrt{v_j}.$$

Exponentiating its endpoints gives an approximate interval for the odds ratio e^{β_j} associated with a one-unit increase in predictor j, with other predictors fixed and without an interaction changing this interpretation. Full column rank and a finite maximum are needed for this calculation; complete or quasi-complete separation can prevent a finite unpenalized logistic maximum.

6.4 Null models, saturated models, and deviance

A null model contains only an intercept, so it assigns the same mean to every observation. A saturated model has enough mean parameters to fit a separate mean to each observation. The saturated model supplies the largest possible log likelihood within the response family; for boundary fits, interpret that value as a supremum.

Let ℓ_{model} denote the maximized log likelihood for a fitted model, with a common fixed dispersion when one is present. Its likelihood ratio deviance is

$$D_{\text{model}} = 2(\ell_{\text{sat}} - \ell_{\text{model}}) \geq 0.$$

It measures discrepancy between the fitted and saturated models. It can often be written as a sum of squared deviance residuals, $D_{\text{model}} = \sum_i d_i^2$.

Exercise 6.3. Find the null-model fit and the saturated log likelihood for independent Bernoulli observations. Derive the deviance of any fitted probabilities $\hat{p}_i \in (0, 1)$.

Solution. For a common probability p, the log likelihood is

$$\ell_0(p) = \left(\sum_i y_i\right) \log p + \left(n - \sum_i y_i\right) \log(1 - p).$$

If $0 < \bar{y} < 1$, differentiation gives the maximizer $\hat{p}_0 = \bar{y}$. If all responses are equal, the maximum is at $p = 0$ or 1; the logit intercept is then infinite in the limiting sense.

For a saturated Bernoulli model take $\hat{p}_i = y_i$. Each observed response then has limiting probability one, so $\ell_{\text{sat}} = 0$, using $0 \log 0 = 0$. Therefore

$$\begin{aligned} D_{\text{model}} &= -2 \sum_i \{y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)\} \\ &= 2 \sum_i \left\{ y_i \log \frac{y_i}{\hat{p}_i} + (1 - y_i) \log \frac{1 - y_i}{1 - \hat{p}_i} \right\}. \end{aligned}$$

For binary observations, a signed square root of each contribution gives

$$d_i = \operatorname{sign}(y_i - \hat{p}_i) \sqrt{2 \left\{ y_i \log \frac{y_i}{\hat{p}_i} + (1 - y_i) \log \frac{1 - y_i}{1 - \hat{p}_i} \right\}},$$

and $D_{\text{model}} = \sum_i d_i^2$.

For normal observations with common known variance σ^2, the saturated means are $\mu_i = y_i$, and

$$2(\ell_{\text{sat}} - \ell_{\text{model}}) = \frac{\sum_i (y_i - \hat{\mu}_i)^2}{\sigma^2}.$$

Thus deviance reduces to residual sum of squares after choosing unit dispersion, or in the convention of an unscaled Gaussian deviance. If σ^2 is estimated separately in each Gaussian model, likelihood-ratio comparisons use the maximized normal likelihood; one cannot treat the variance as a common known constant or maximize a zero-residual saturated normal model over an unrestricted positive variance.

6.4.1 Comparing nested models

Suppose the reduced model is contained in the full model, with parameter dimensions $q_{\mathrm{R}} < q_{\mathrm{F}}$. Using the same saturated model and dispersion convention,

$$\begin{aligned}\Delta D &= D_{\mathrm{R}} - D_{\mathrm{F}} \\ &= 2(\ell_{\mathrm{F}} - \ell_{\mathrm{R}}) \geq 0.\end{aligned}$$

Under the null hypothesis that the reduced model is correct and the usual likelihood regularity conditions,

$$\Delta D \;\dot{\sim}\; \chi^2_{q_{\mathrm{F}} - q_{\mathrm{R}}}$$

for large samples. The degrees of freedom equal the number of additional free parameters, also the difference between the residual degrees of freedom of the two models. This is a test of the added terms after allowing for the terms already in the reduced model. Boundary parameters, nonidentifiability, or a nonexistent finite maximum can invalidate this approximation.

6.5 Information criteria

Likelihood fit improves when more parameters are available. Information criteria balance fit against model dimension. If q is the number of estimated parameters and n is the number of independent observations,

$$\mathrm{AIC} = -2\ell_{\mathrm{model}} + 2q, \qquad \mathrm{BIC} = -2\ell_{\mathrm{model}} + q \log n.$$

AIC is the Akaike information criterion; BIC is the Bayesian information criterion. Smaller values are preferred. Count all fitted parameters, including a dispersion parameter when estimated. Compare likelihoods for the same response data and compatible probability models. A larger model must improve its log likelihood sufficiently to compensate for its larger penalty.

Chapter 7

Cross-Validation and Prediction Error

7.1 Resampling and the validation-set approach

Resampling repeatedly reuses the available observations to assess a method. Cross-validation estimates prediction accuracy; the bootstrap can assess estimation accuracy, such as the variability of an estimator.

In the validation-set approach, randomly split the observations into a training set T and a validation set V. Fit the method using only T and call the resulting function $\hat{f}_T$. For a quantitative response, compute

$$\widehat{\mathrm{MSE}}_V = \frac{1}{|V|} \sum_{i \in V} \{y_i - \hat{f}_T(x_i)\}^2.$$

This estimates the test MSE of the procedure trained on T. Candidate methods or different degrees of flexibility can be compared using their validation MSE, rather than their training MSE.

The distinction between these errors is essential. If $\hat{f} = \hat{f}((x_1, y_1), \ldots, (x_n, y_n))$, then a training observation is also an input to the fitted rule. Consequently,

$$\frac{1}{n} \sum_i \{y_i - \hat{f}(x_i)\}^2$$

is not an average of independent test losses for this rule. Fitting and assessment on the same observations usually yields an optimistic estimate of prediction error. The validation observations must instead represent new observations from the same target distribution. With independent identically distributed observations, a random split supplies this separation.

The validation-set approach sets aside a substantial portion of the data. It has two drawbacks: the result can vary markedly with the random split, and the fitted method uses fewer observations than the final model trained on all n. For procedures that improve with additional training data, this can overestimate the prediction error of the final fit.

7.2 Leave-one-out cross-validation

For each i, omit observation (x_i, y_i) and fit the method to the remaining $n - 1$ observations. Let $\hat{f}^{(-i)}$ be this fitted function. Predict the omitted response and compute

$$\mathrm{MSE}_i = \{y_i - \hat{f}^{(-i)}(x_i)\}^2.$$

The leave-one-out cross-validation estimate is

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\mathrm{MSE}_i.$$

Each fit uses almost the whole sample, and each observation is assessed using a model that did not use its response. LOOCV does not require a random choice of folds, but fitting n models can be computationally intensive.

7.2.1 A shortcut for ordinary least squares

For a fixed full-rank design, let

$$H = X(X^TX)^{-1}X^T, \qquad \hat{Y} = HY, \qquad h_i = H_{ii}, \qquad e_i = y_i - \hat{y}_i.$$

Assume deleting any observation leaves the model identifiable, so $h_i < 1$. Then ordinary least squares has the shortcut

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1-h_i}\right)^2.$$

Thus only the original fit is needed. The residuals are scaled by leverage; this differs from simply averaging their squares in the training sample.

Exercise 7.1. Derive the identity $y_i - \hat{f}^{(-i)}(x_i) = e_i/(1-h_i)$.

Solution. Let x_i be the column design vector for observation i, $A = X^TX$, and $b = X^TY$. The full fit is $\hat{\beta} = A^{-1}b$. Deleting observation i changes the normal equations to

$$(A - x_ix_i^T)\hat{\beta}^{(-i)} = b - x_iy_i.$$

Subtract $(A - x_ix_i^T)\hat{\beta} = b - x_i\hat{y}_i$ to obtain

$$(A - x_ix_i^T)(\hat{\beta}^{(-i)} - \hat{\beta}) = -x_ie_i.$$

Since $h_i = x_i^TA^{-1}x_i$, the identity

$$(A - x_ix_i^T)A^{-1}x_i = x_i(1-h_i)$$

gives

$$\hat{\beta}^{(-i)} - \hat{\beta} = -\frac{A^{-1}x_ie_i}{1-h_i}.$$

Multiplying by x_i^T,

$$\hat{f}^{(-i)}(x_i) = \hat{y}_i - \frac{h_ie_i}{1-h_i}.$$

Therefore

$$y_i - \hat{f}^{(-i)}(x_i) = e_i + \frac{h_ie_i}{1-h_i} = \frac{e_i}{1-h_i}.$$

Squaring and averaging proves the shortcut. The calculation assumes the same fixed model is fit each time; a procedure that repeats variable selection inside every training set requires assessing that whole procedure.

7.3 k-fold cross-validation

Randomly divide the data into k disjoint folds $V_1, \ldots, V_k$ of approximately equal sizes. For fold r, fit on all observations outside V_r and evaluate on V_r:

$$\mathrm{MSE}_r = \frac{1}{|V_r|} \sum_{i \in V_r} \{y_i - \hat{f}^{(-r)}(x_i)\}^2.$$

For equal fold sizes, the cross-validation estimate is

$$\mathrm{CV}_{(k)} = \frac{1}{k} \sum_{r=1}^{k} \mathrm{MSE}_r.$$

For unequal sizes, weight by the number of evaluated observations:

$$\mathrm{CV}_{(k)} = \sum_{r=1}^{k} \frac{|V_r|}{n} \mathrm{MSE}_r = \frac{1}{n} \sum_{r=1}^{k} \sum_{i \in V_r} \{y_i - \hat{f}^{(-r)}(x_i)\}^2.$$

This fits the model only k times. Common choices are $k = 5$ and $k = 10$; LOOCV is the special case $k = n$. Each fold plays the role of the omitted validation set once.

7.3.1 Bias and variance of the error estimate

The validation-set estimate can have upward bias for the error of the final model because its training set is smaller. The same issue remains for k-fold CV, but less severely when k is larger and each training set contains more observations. Under independent identical sampling, LOOCV is unbiased for the expected prediction error of the procedure trained on $n - 1$ observations. It is not, in general, exactly unbiased for the risk of a fit trained on all n observations.

Bias alone does not determine the quality of the estimate. If Z_i denotes one evaluated loss, then

$$\mathrm{Var}\left(\frac{1}{m} \sum_{i=1}^{m} Z_i\right) = \frac{1}{m^2} \left\{ \sum_i \mathrm{Var}(Z_i) + 2 \sum_{i<j} \mathrm{Cov}(Z_i, Z_j) \right\}.$$

The LOOCV fits use highly overlapping training sets. Their losses need not be independent, and positive correlations can keep the variance of their average large. The fits in k-fold CV differ more substantially, which can reduce these correlations. Consequently, 5- or 10-fold CV often has lower variance than LOOCV, though this is not a universal ordering for every method and dataset. This is the bias–variance trade-off for the error estimate itself.

7.4 Cross-validation for classification

Use exactly the same fitting and evaluation scheme, replacing squared error by zero–one loss. If $\hat{g}^{(-r)}$ is the classifier fit without fold r, let

$$\mathrm{Err}_r = \frac{1}{|V_r|} \sum_{i \in V_r} \mathbf{1}\{y_i \neq \hat{g}^{(-r)}(x_i)\}.$$

For equal-sized folds,

$$\mathrm{CV}_{(k)} = \frac{1}{k}\sum_{r=1}^{k} \mathrm{Err}_r;$$

otherwise use the weights $|V_r|/n$. The estimated quantity is a misclassification rate rather than a mean squared error. Validation and future test observations must come from the same target distribution for this error to assess the intended prediction problem.

All preprocessing steps that are estimated from data, including predictor scaling and variable selection, must be fitted within each training fold and then applied to its held-out fold. When cross-validation selects a tuning parameter, a separate test set can assess the selected procedure.

Chapter 8

Model Selection, Regularization, and Dimension Reduction

8.1 Selecting a Subset of Predictors

Consider a linear model with p available predictors. Three related approaches are variable selection, shrinkage or regularization, and dimension reduction. Variable selection uses a subset of the original predictors. Shrinkage controls the sizes of their coefficients. Dimension reduction replaces the predictors by a smaller set of combinations.

For observations (x_i, y_i), write

$$\mathrm{RSS}(\beta_0, \beta) = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j \right)^2 .$$

An intercept is included throughout and is not counted among the p predictors.

8.1.1 Best Subset Selection

There are

$$\sum_{k=0}^{p} \binom{p}{k} = 1 + p + \binom{p}{2} + \cdots + \binom{p}{p-1} + 1 = 2^p$$

possible subsets. Best subset selection proceeds as follows.

1. Let M_0 be the null model, containing only an intercept.
2. For each $k = 1, \ldots, p$, fit all $\binom{p}{k}$ models with k predictors. Among these models, retain the one with smallest RSS, equivalently largest training R^2, and call it M_k.
3. Compare $M_0, \ldots, M_p$ using an estimate of test error or a criterion that accounts for model size. Possible choices include cross-validation, AIC, BIC, and adjusted R^2.

For a fixed budget s, the search for a model with at most s predictors can be written

$$\min_{\beta_0, \beta} \mathrm{RSS}(\beta_0, \beta) \quad \text{subject to} \quad \sum_{j=1}^{p} \mathbf{1}\{\beta_j \neq 0\} \leq s.$$

The constraint counts nonzero slope coefficients. Solving this problem for the possible sizes produces the candidate models; a separate comparison selects a size. Searching 2^p subsets is computationally expensive, and searching a very large collection can also overfit the training data.

8.1.2 Stepwise Selection

Forward stepwise selection begins with M_0. Given a model M_k with k predictors, fit each of the $p-k$ models obtained by adding one unused predictor. Retain the best of these models as M_{k+1}. Repeat for $k = 0, \dots, p-1$, and then compare the resulting candidates using a test-error estimate or a model-selection criterion.

Backward stepwise selection begins with the full model. At each step, consider deleting each currently included predictor, retain the deletion that gives the best reduced model, and continue down to the null model. The full least-squares fit must be well defined for the usual backward procedure; this is a restriction when the number of predictors is large relative to n.

A *hybrid* procedure allows additions and deletions. For example, after adding a predictor it may reconsider whether a previously included predictor should be removed.

Example 8.1 (Three candidate predictors)**.** Let the candidates be X_1, X_2, X_3. Begin with

$$M_0 : \quad f(x) = \beta_0.$$

The first candidates are

$$\beta_0 + \beta_1 x_1, \qquad \beta_0 + \beta_2 x_2, \qquad \beta_0 + \beta_3 x_3.$$

Suppose the first has the largest fitted R^2. Then M_1 uses X_1. At the second step compare

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2, \qquad \beta_0 + \beta_1 x_1 + \beta_3 x_3.$$

Suppose the second is better. The retained model is

$$M_2 : \quad f(x) = \beta_0 + \beta_1 x_1 + \beta_3 x_3.$$

The last addition gives the full model

$$M_3 : \quad f(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3.$$

Choose among M_0, M_1, M_2, M_3 using a criterion that accounts for model size, such as cross-validation, adjusted R^2, AIC, or BIC. The full model's largest training R^2 is not a reason by itself to select it.

For backward selection, a possible first deletion is

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 \quad \longrightarrow \quad \beta_0 + \beta_2 x_2 + \beta_3 x_3.$$

These are greedy, path-dependent searches. They optimize the choice at the current step and are not guaranteed to find the best subset among all 2^p possibilities. Forward and backward paths need not agree, and the candidates of a given size need not equal the corresponding best-subset candidate.

8.1.3 Dimension relative to sample size

High dimensionality is relative to the number of observations. For example,

$$n = 10, \quad p = 30$$

is high-dimensional for ordinary unregularized regression, whereas

$$n = 1000, \quad p = 30$$

is a much less demanding ratio of predictor dimension to sample size. In particular, a design with $p + 1$ columns cannot have full column rank if $p + 1 > n$. Even when $p + 1 \leq n$, correlations among predictors can make coefficient estimation unstable.

8.2 Comparing Models of Different Sizes

Training RSS alone cannot select a model size: enlarging a model cannot increase its least-squares training RSS. We need a measure of performance on new observations.

A validation set or cross-validation estimates test error directly by evaluating predictions on observations excluded from fitting. Another approach adjusts training error to account for fitting a model to the same data on which its error is measured.

Let d be the number of included predictors, so that $q = d + 1$ mean parameters are fitted when an intercept is present. Suppose $n > d + 1$ and define

$$\mathrm{TSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2.$$

The adjusted coefficient of determination is

$$R^2_{\mathrm{adj}} = 1 - \frac{\mathrm{RSS}\,/(n - d - 1)}{\mathrm{TSS}\,/(n - 1)}.$$

Larger adjusted R^2 is preferred. Unlike training R^2, it can decrease when an additional predictor does not improve the fit sufficiently.

For homoskedastic linear regression, a prediction-error version of Mallows' criterion is

$$\frac{\mathrm{RSS} + 2q\widehat{\sigma}^2}{n}.$$

Here $\widehat{\sigma}^2$ is a common estimate of the error variance, for example from an adequately specified larger model. This is the training MSE plus a complexity correction. It differs from the traditional dimensionless

$$C_p = \frac{\mathrm{RSS}}{\widehat{\sigma}^2} - n + 2q$$

by a positive scale and a model-independent constant. With a common intercept, using d instead of q in the penalty also changes all candidate criteria by the same constant.

Recall the likelihood definitions of AIC and BIC. For Gaussian regression evaluated with a common fixed variance scale $\widehat{\sigma}^2$, equivalent normalized comparison criteria are

$$\frac{\mathrm{RSS} + 2q\widehat{\sigma}^2}{n\widehat{\sigma}^2}, \qquad \frac{\mathrm{RSS} + q\widehat{\sigma}^2 \log n}{n\widehat{\sigma}^2}.$$

These expressions use a common variance; they are not the Gaussian likelihood with a separately re-estimated variance in each model. In the common-variance comparison, AIC and the prediction-error version of C_p rank models identically. Smaller values are preferred. For $\log n > 2$, BIC penalizes the number of fitted parameters more heavily than AIC.

For logistic regression and other generalized linear models, deviance can compare fit among models with the same size. Validation error, cross-validation, AIC, or BIC can compare different sizes. The adjusted-R^2 and RSS formulas above are specifically linear-regression formulas.

8.3 Shrinkage and Regularization

Shrinkage fits a model using the available predictors while pulling slope coefficients toward zero. Ridge regression uses an ℓ_2 penalty; the lasso uses an ℓ_1 penalty. For $\beta \in \mathbb{R}^p$,

$$\|\beta\|_1 = \sum_{j=1}^{p} |\beta_j|, \qquad \|\beta\|_2 = \left(\sum_{j=1}^{p} \beta_j^2 \right)^{1/2}.$$

These methods are especially useful when fitting many predictors causes high variance or instability.

8.3.1 Least Squares, Centering, and Scaling

Under the linear model

$$Y = \beta_0 \mathbf{1} + X\beta + \varepsilon, \qquad \mathbb{E}(\varepsilon \mid X) = 0, \qquad \operatorname{Var}(\varepsilon \mid X) = \sigma^2 I_n,$$

ordinary least squares is unbiased when the design has full column rank. If the predictor columns are centered, $\mathbf{1}^T X = 0$, the intercept estimate is $\widehat{\beta}_0 = \bar{y}$ and the slope estimate is

$$\widehat{\beta} = (X^T X)^{-1} X^T Y, \qquad \operatorname{Cov}(\widehat{\beta} \mid X) = \sigma^2 (X^T X)^{-1}.$$

Centering therefore lets us handle the intercept separately rather than adding a column of ones to X.

If a predictor is multiplied by a nonzero constant c, its ordinary least-squares slope is divided by c, and its contribution $\widehat{\beta}_j x_j$ to a fitted value remains unchanged. This is scale equivariance.

Highly correlated predictors make $X^T X$ nearly singular. Its inverse can then be large in some directions, producing high-variance estimates. If X is rank deficient, ordinary least-squares coefficients are not unique; the displayed inverse and covariance formula do not apply. In particular, a centered design with too many columns cannot have full column rank. This is different from claiming that a uniquely defined ordinary least-squares estimator has an infinite variance.

8.3.2 Ridge Regression

Ridge regression solves

$$\min_{\beta_0, \beta} \operatorname{RSS}(\beta_0, \beta) \quad \text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 \le s, \qquad s \ge 0.$$

Its penalized form is

$$\min_{\beta_0,\beta}\left\{\mathrm{RSS}(\beta_0,\beta)+\lambda\sum_{j=1}^{p}\beta_j^2\right\},\qquad \lambda\geq 0.$$

The penalty is $\lambda\|\beta\|_2^2$ and applies only to the slopes. The intercept is unpenalized. The budget s controls how large the coefficient vector may be; the multiplier λ balances fit against coefficient size. Larger budgets correspond to weaker shrinkage. Constrained and penalized solutions can be matched through their multipliers, although there need not be a strictly one-to-one relationship at inactive constraints or degenerate solutions.

Deriving the ridge estimate. Assume the columns of X are centered and $\lambda>0$. Differentiating the objective with respect to the unpenalized intercept gives

$$-2\mathbf{1}^T(Y-\beta_0\mathbf{1}-X\beta)=0.$$

Since $\mathbf{1}^TX=0$, we obtain $\widehat{\beta}_0^R=\bar{y}$. With $Y_c=Y-\bar{y}\mathbf{1}$, the slope objective has the quadratic expansion

$$Y_c^TY_c-2\beta^TX^TY+\beta^T(X^TX+\lambda I_p)\beta.$$

Differentiating with respect to β gives

$$-2X^T(Y-\bar{y}\mathbf{1}-X\beta)+2\lambda\beta=0.$$

Again $X^T\mathbf{1}=0$, so

$$(X^TX+\lambda I_p)\widehat{\beta}^R=X^TY,\qquad \widehat{\beta}^R=(X^TX+\lambda I_p)^{-1}X^TY.$$

In the original uncentered predictor coordinates, the intercept is $\bar{y}-\sum_j\bar{x}_j\widehat{\beta}_j^R$.

Exercise 8.1. Show that the ridge inverse is well defined for $\lambda>0$, including for a rank-deficient design. Compute the conditional bias and covariance under the stated linear model, compare its variance with least squares, and find its limit as $\lambda\to\infty$.

Solution. For any nonzero v and $\lambda>0$,

$$v^T(X^TX+\lambda I_p)v=\|Xv\|_2^2+\lambda\|v\|_2^2>0.$$

Thus the matrix is positive definite and invertible. Adding a positive constant to its diagonal stabilizes the inversion.

The estimate is linear in Y. Under the stated linear model,

$$\begin{aligned}\mathbb{E}(\widehat{\beta}^R\mid X)&=(X^TX+\lambda I_p)^{-1}X^TX\beta,\\ \mathbb{E}(\widehat{\beta}^R\mid X)-\beta&=-\lambda(X^TX+\lambda I_p)^{-1}\beta,\\ \mathrm{Cov}(\widehat{\beta}^R\mid X)&=\sigma^2(X^TX+\lambda I_p)^{-1}X^TX(X^TX+\lambda I_p)^{-1}.\end{aligned}$$

Consequently ridge is generally biased. If u is a unit eigenvector of X^TX with eigenvalue $a>0$, then the ordinary least-squares variance in direction u is σ^2/a, while the ridge variance is

$$\sigma^2\frac{a}{(a+\lambda)^2}\leq\frac{\sigma^2}{a}.$$

The expected coefficient in that direction is multiplied by $a/(a+\lambda)$. Ridge trades bias for reduced variance, particularly in directions with small eigenvalues.

To see the limiting behavior, diagonalize $X^TX = V\,\mathrm{diag}(d_1,\dots,d_p)V^T$, with $d_j \geq 0$. The coefficient of v_j in the expectation is multiplied by $d_j/(d_j+\lambda)$, which tends to zero as $\lambda \to \infty$. For the estimate itself,

$$\widehat{\beta}^R = V\,\mathrm{diag}\{(d_j+\lambda)^{-1}\}V^TX^TY \longrightarrow 0.$$

With an unpenalized intercept, the limiting fitted values are all $\bar{y}$. The amount of shrinkage is controlled by the tuning parameter λ.

When the ordinary least-squares estimate exists, $\lambda = 0$ gives that estimate. As $\lambda \to \infty$, all slopes tend to zero and the fit tends to the intercept-only model. With rank deficiency, the limit as $\lambda \downarrow 0$ is the minimum-norm least-squares slope solution.

Exercise 8.2. Show that the Euclidean norm of the ridge slope estimate is nonincreasing as λ increases.

Solution. Let $\lambda_2 > \lambda_1 \geq 0$ and let b_1, b_2 minimize the corresponding penalized objectives. Set $R(b)$ equal to the RSS after fitting the intercept. Optimality gives

$$R(b_1) + \lambda_1 \|b_1\|_2^2 \leq R(b_2) + \lambda_1 \|b_2\|_2^2,$$
$$R(b_2) + \lambda_2 \|b_2\|_2^2 \leq R(b_1) + \lambda_2 \|b_1\|_2^2.$$

Adding and rearranging yields

$$(\lambda_2 - \lambda_1)(\|b_2\|_2^2 - \|b_1\|_2^2) \leq 0.$$

Therefore $\|b_2\|_2 \leq \|b_1\|_2$. The statement also holds when the norm stays constant.

Ridge generally retains the original predictors rather than making an exactly sparse selection, though individual coefficients can equal zero in particular data sets. The penalty is not scale equivariant: multiplying a predictor by c and dividing its coefficient by c changes its penalty contribution by a factor $1/c^2$. Predictors measured on different scales should therefore be standardized when equal shrinkage treatment is intended.

Increasing λ strengthens shrinkage, generally increasing squared bias while reducing variance. Ridge is useful when the reduction in variance outweighs the increase in squared bias, for example with strongly correlated predictors or when p is comparable to or exceeds n. The tuning parameter controls this bias–variance tradeoff and can be chosen using validation or cross-validation.

8.3.3 The Lasso

The lasso, or least absolute shrinkage and selection operator, solves

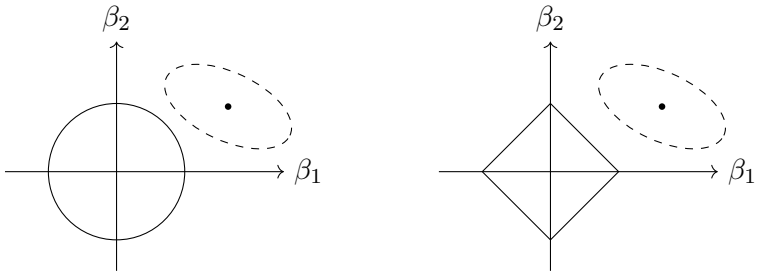
$$\min_{\beta_0,\beta} \mathrm{RSS}(\beta_0,\beta) \quad \text{subject to} \quad \sum_{j=1}^{p} |\beta_j| \leq s.$$

Its penalized objective is

$$\mathrm{RSS}(\beta_0,\beta) + \lambda \sum_{j=1}^{p} |\beta_j|.$$

As with ridge, the intercept is unpenalized and each value of λ gives a fitted coefficient vector. The ℓ_1 penalty can set coefficients exactly to zero, so the lasso performs variable selection as well as shrinkage. In general its estimates are not linear functions of Y, and there is no single matrix-inverse formula analogous to the ridge formula. The lasso also depends on predictor scales.

The constrained formulations give a geometric interpretation. RSS has ellipsoidal contours when the design has full column rank. The constrained solution occurs where the smallest attainable contour first meets the constraint region. In two dimensions the ridge region is a disk and the lasso region is a diamond. The diamond's corners lie on coordinate axes, making contact at a zero coefficient possible.



Ridge: $\beta_1^2 + \beta_2^2 \le s$ Lasso: $|\beta_1| + |\beta_2| \le s$

The dashed curves illustrate RSS contours, and the points indicate unconstrained estimates. Their positions are schematic.

8.4 Principal Components and Dimension Reduction

Principal component regression and partial least squares replace the original predictors by a smaller number of linear combinations. Principal component regression uses principal components computed from X without using Y. Partial least squares also constructs predictor combinations, but uses information about their relationship with Y. Thus PCA itself is an unsupervised method; partial least squares is response-informed.

8.4.1 Loadings and Scores

PCA transforms p possibly correlated variables into p uncorrelated principal components, ordered by decreasing sample variance. The first component has the greatest variance among unit-length linear combinations. The second maximizes variance among unit-length combinations uncorrelated with the first, and so on.

Let X be an $n \times p$ matrix whose columns have sample mean zero. For simplicity suppose $p < n$, though the spectral construction also allows zero eigenvalues. The sample covariance matrix is

$$S = \frac{X^T X}{n-1}.$$

It is symmetric positive semidefinite. Let (λ_k, ϕ_k) be its eigenvalue–eigenvector pairs, ordered so that

$$S\phi_k = \lambda_k \phi_k, \qquad \lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_p \ge 0.$$

Choose the eigenvectors orthonormally:

$$\phi_k^T \phi_k = 1, \qquad \phi_k^T \phi_\ell = 0 \quad (k \neq \ell).$$

Writing $\Phi = (\phi_1, \dots, \phi_p)$ and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_p)$ gives the spectral decomposition

$$S = \Phi \Lambda \Phi^T, \qquad \Phi^T \Phi = \Phi \Phi^T = I_p.$$

For a centered predictor vector $x = (x_1, \dots, x_p)^T$, the kth principal component is

$$Z_k = \phi_k^T x = \phi_{1k} x_1 + \cdots + \phi_{pk} x_p.$$

The coefficients ϕ_{jk} are its *loadings*. Evaluating this combination at observation i gives the *score*

$$z_{ik} = \phi_{1k} x_{i1} + \cdots + \phi_{pk} x_{ip}.$$

The score matrix is

$$Z = X\Phi, \qquad Z\Phi^T = X.$$

Thus retaining all components is an orthogonal change of coordinates, not a loss of information.

Proposition 8.2. *The score columns have mean zero and sample covariance matrix Λ. In particular they are pairwise uncorrelated and the variance of component k is λ_k.*

Proof. Since $\sum_i x_i = 0$,

$$\bar{z}_k = \frac{1}{n} \sum_{i=1}^n x_i^T \phi_k = \left(\frac{1}{n} \sum_{i=1}^n x_i \right)^T \phi_k = 0.$$

The sample covariance matrix is

$$\frac{Z^T Z}{n-1} = \Phi^T \frac{X^T X}{n-1} \Phi = \Phi^T S \Phi = \Phi^T \Phi \Lambda \Phi^T \Phi = \Lambda.$$

The off-diagonal entries are zero and the diagonal entries are λ_k.

Exercise 8.3. Find the sample covariance between component Z_k and original predictor X_j. When both sample variances are positive, find their sample correlation.

Solution. Let e_j be the jth coordinate vector, so that X_j has score column Xe_j. Then

$$\widehat{\operatorname{Cov}}(Z_k, X_j) = \frac{(X\phi_k)^T (Xe_j)}{n-1} = \phi_k^T S e_j = (S\phi_k)^T e_j = \lambda_k \phi_{jk}.$$

Since $\widehat{\operatorname{Var}}(Z_k) = \lambda_k$ and $\widehat{\operatorname{Var}}(X_j) = S_{jj}$,

$$\widehat{\operatorname{Corr}}(Z_k, X_j) = \frac{\lambda_k \phi_{jk}}{\sqrt{\lambda_k S_{jj}}} = \frac{\sqrt{\lambda_k} \phi_{jk}}{\sqrt{S_{jj}}}.$$

The correlation is undefined if either variable has zero sample variance.

8.4.2 The Variance Maximization Problem

A linear combination $a^T x$ has sample mean zero and sample variance $a^T S a$. Without a constraint on a, this variance can be made arbitrarily large by rescaling any combination with positive variance. PCA therefore imposes $a^T a = 1$.

Exercise 8.4. Solve

$$\max_{a \in \mathbb{R}^p} a^T S a \quad \text{subject to} \quad a^T a = 1,$$

and then solve the same problem subject to orthogonality to the preceding maximizing directions.

Solution. Write $a = \Phi c$. Since Φ is orthogonal,

$$a^T a = c^T c = \sum_{j=1}^{p} c_j^2 = 1, \qquad a^T S a = c^T \Lambda c = \sum_{j=1}^{p} \lambda_j c_j^2 \leq \lambda_1.$$

Equality is attained by $c = e_1$, hence by $a = \phi_1$. More generally any unit vector in the largest-eigenvalue eigenspace maximizes the variance.

For the kth direction, impose $a^T \phi_j = 0$ for $j < k$. These constraints give $c_j = 0$ for $j < k$, and therefore

$$a^T S a = \sum_{j=k}^{p} \lambda_j c_j^2 \leq \lambda_k \sum_{j=k}^{p} c_j^2 = \lambda_k.$$

Equality is attained at $a = \phi_k$. The eigenvectors can therefore be selected successively in descending order of variance. Repeated eigenvalues allow more than one orthonormal choice within an eigenspace.

For principal component regression, retain $m \leq p$ score columns and fit a regression on those scores:

$$y_i \approx \theta_0 + \sum_{k=1}^{m} \theta_k z_{ik}.$$

This restricts the original slope vector to the span of $\phi_1, \ldots, \phi_m$. The reduction uses directions with large predictor variance; because the PCA step does not use the response, large predictor variance alone does not guarantee strong predictive value.

Chapter 9

Beyond Linearity

9.1 Expanding the Predictor Space

In regression we model $f(x) = \mathbb{E}(Y \mid X = x)$. For a binary response we model $p(x) = \mathbb{P}(Y = 1 \mid X = x)$. Ordinary linear regression and logistic regression assume, respectively,

$$f(x) = \beta_0 + \sum_{j=1}^{p} \beta_j x_j, \qquad \log \frac{p(x)}{1 - p(x)} = \beta_0 + \sum_{j=1}^{p} \beta_j x_j.$$

These expressions are linear in the predictors. A model can remain linear in its unknown coefficients while allowing a nonlinear relationship with x. Useful constructions include polynomials, step functions, splines, and local regression. Generalized additive models extend the one-predictor constructions to several predictors.

9.1.1 Polynomial Regression

With one predictor, a degree-d polynomial model is

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_d x^d,$$

or, for logistic regression,

$$\log \frac{p(x)}{1 - p(x)} = \beta_0 + \beta_1 x + \cdots + \beta_d x^d.$$

The added predictors are $x, x^2, \ldots, x^d$. The coefficients still enter linearly, so least squares applies in the regression setting and maximum likelihood applies in the logistic setting. This expands the feature space without changing the basic fitting principle. The degree can be selected by cross-validation or by testing nested polynomial models. Low degrees, often no greater than three, avoid the instability that can arise with a high-degree polynomial over a large range.

9.1.2 Step Functions

Choose cutpoints $c_1 < \cdots < c_K$ to divide the predictor range into $K + 1$ bins. Define

$$\begin{aligned} C_0(x) &= \mathbf{1}\{x < c_1\}, \\ C_j(x) &= \mathbf{1}\{c_j \le x < c_{j+1}\}, \qquad j = 1, \ldots, K - 1, \\ C_K(x) &= \mathbf{1}\{c_K \le x\}. \end{aligned}$$

A piecewise-constant model can be written with the first bin as the reference category:

$$g(x) = \beta_0 + \sum_{j=1}^{K} \beta_j C_j(x),$$

where $g = f$ in regression and $g = \log\{p/(1-p)\}$ in logistic regression. In the first bin, $g(x) = \beta_0$. In bin $j \geq 1$, $g(x) = \beta_0 + \beta_j$, so β_j is the difference from the reference-bin value. Including all $K+1$ bin indicators together with an intercept would make the design linearly dependent because the indicators sum to one.

9.1.3 Basis Functions

More generally, choose known functions $b_1(x), \ldots, b_K(x)$ and fit

$$g(x) = \beta_0 + \sum_{k=1}^{K} \beta_k b_k(x).$$

Polynomials and step functions are special cases. Other choices include Fourier functions, wavelets, and spline bases. Once the basis is specified, its values form the columns of a design matrix and the unknown coefficients can be estimated by the appropriate regression procedure.

9.2 Regression Splines

A regression spline fits separate polynomials over intervals and joins them at *knots*. For a single knot c, an unconstrained piecewise-cubic model is

$$g(x) = \begin{cases} a_0 + a_1 x + a_2 x^2 + a_3 x^3, & x < c, \\ d_0 + d_1 x + d_2 x^2 + d_3 x^3, & x \geq c. \end{cases}$$

There are initially eight coefficients. A smooth cubic spline requires continuity of g, g', and g'' at c:

$$g(c^-) = g(c^+), \quad g'(c^-) = g'(c^+), \quad g''(c^-) = g''(c^+).$$

Each condition removes one degree of freedom. With K distinct interior knots, a cubic spline has $4 + K$ free coefficients before any additional boundary conditions. Linear and quadratic splines use the corresponding lower-degree pieces and continuity requirements.

Exercise 9.1. Verify the degrees of freedom of a cubic spline with K knots and construct a basis for it.

Solution. There are $K+1$ intervals, each with four cubic coefficients, giving $4(K+1)$ parameters. At each interior knot, matching the function and its first two derivatives imposes three independent constraints. Hence the dimension is

$$4(K+1) - 3K = K + 4.$$

Define $(u)_+ = \max(u, 0)$. For distinct knots $c_1, \ldots, c_K$, use

$$1, \quad x, \quad x^2, \quad x^3, \quad (x - c_1)_+^3, \ldots, (x - c_K)_+^3.$$

Each truncated cubic is zero to the left of its knot and cubic to the right. Its value and first two derivatives agree at the knot, while its third derivative jumps by six. Given a cubic spline, subtract its polynomial on the first interval. At the first knot, matching through the second derivative forces the change of polynomial to be a multiple of $(x - c_1)^3$. Subtract that multiple of $(x - c_1)^3_+$ and repeat at the remaining knots. This represents the spline in the displayed basis. The distinct third-derivative jumps also show independence of the truncated terms; the four ordinary polynomials are independent as well.

9.3 Smoothing Splines

Minimizing RSS over unrestricted functions can merely interpolate the observations. Exact interpolation may fit noise and need not give a useful smooth curve. A smoothing spline balances fit with a roughness penalty:

$$\min_g \left\{ \sum_{i=1}^{n} \big(y_i - g(x_i)\big)^2 + \lambda \int_a^b \big(g''(t)\big)^2 \, dt \right\}, \qquad \lambda \geq 0,$$

where $[a, b]$ spans the observed predictor values and the candidate functions have square-integrable second derivatives. The integral measures overall curvature. A line $g(t) = \beta_0 + \beta_1 t$ has $g'' = 0$ and therefore zero roughness penalty.

Assume there are at least two distinct predictor values. At $\lambda = 0$ there is no roughness penalty, and interpolation is possible when the predictor values are distinct. For $\lambda > 0$, the minimizing function is a natural cubic spline with potential knots at the distinct observed x_i. Natural boundary conditions make it linear outside the boundary knots. Increasing λ suppresses curvature; as $\lambda \to \infty$, the fit tends to the least-squares line, assuming at least two distinct predictor values.

A natural cubic regression spline and a smoothing spline use related function spaces, but their fitting choices differ. A regression spline typically specifies a smaller collection of knots and fits the coefficients without a roughness penalty. A smoothing spline uses the observed predictor values as potential knots and selects smoothness through λ.

Exercise 9.2. Explain why the roughness-penalized minimizer can be taken to be a natural cubic spline. Then derive the matrix form of the fit and its effective degrees of freedom.

Solution. Assume first that $a = x_1 < \cdots < x_n = b$ are distinct. For any candidate g, let s be the natural cubic spline interpolating its values at these points. Put $h = g - s$, so that $h(x_i) = 0$ for every i. On each interval (x_i, x_{i+1}), s is cubic and $s^{(4)} = 0$. Integrating by parts twice on all intervals gives

$$\int_a^b s''h'' = \sum_{i=1}^{n-1} \big[\, s''h' - s'''h \,\big]_{x_i}^{x_{i+1}} = 0.$$

Indeed, the terms containing $h(x_i)$ vanish; the interior terms containing h' cancel because s'' is continuous; and the boundary terms vanish because $s''(a) = s''(b) = 0$. Thus

$$\int_a^b (g'')^2 = \int_a^b (s'')^2 + \int_a^b (h'')^2 \geq \int_a^b (s'')^2.$$

The interpolation values, hence the RSS, are unchanged. Therefore minimizing over natural cubic splines is sufficient. The natural cubic interpolant exists uniquely for distinct nodes: its piecewise-cubic coefficients solve the value-matching, derivative-matching, and natural-boundary equations.

Uniqueness also follows by applying the same integration identity to a natural cubic spline with all nodal values zero: its integrated squared second derivative is zero, so it is linear and hence zero. The resulting square linear system is therefore invertible.

Let $b_1, \dots, b_m$ be a basis of this natural spline space, let $B_{ij} = b_j(x_i)$, and write $g(x) = \sum_j \theta_j b_j(x)$. Define

$$\Omega_{jk} = \int_a^b b_j''(t) b_k''(t)\, dt.$$

The objective becomes

$$(Y - B\theta)^T (Y - B\theta) + \lambda \theta^T \Omega \theta.$$

Differentiation gives

$$(B^T B + \lambda \Omega)\widehat{\theta} = B^T Y,$$

so, in a full-rank representation,

$$\widehat{Y} = S_\lambda Y, \qquad S_\lambda = B(B^T B + \lambda \Omega)^{-1} B^T.$$

For the natural interpolation basis at distinct observed values, B is invertible, so this inverse exists for $\lambda \geq 0$. Repeated predictor values can first be grouped with their observation counts as weights.

The matrix S_λ is the $n \times n$ smoothing matrix. Since shrinkage changes the amount of fitting, merely counting basis coefficients does not measure model flexibility. The effective degrees of freedom are

$$\mathrm{df}_\lambda = \mathrm{tr}(S_\lambda).$$

To see their range, use the nodal values as coordinates, so $B = I_n$ and $S_\lambda = (I_n + \lambda \Omega)^{-1}$. The positive semidefinite roughness matrix has a two-dimensional null space consisting of nodal values of lines. If its remaining eigenvalues are $\rho_3, \dots, \rho_n > 0$, then

$$\mathrm{df}_\lambda = 2 + \sum_{j=3}^{n} \frac{1}{1 + \lambda \rho_j}.$$

It decreases from n at zero penalty toward two for a line. This also shows directly why increasing the penalty suppresses curvature.

A smoothing spline is a nonparametric estimate even though, for a fixed data set, its fitted values have a finite matrix representation. The number of potential knots need not be chosen separately. One may choose λ by leave-one-out cross-validation or specify a desired effective degrees of freedom, $\mathrm{tr}(S_\lambda)$. Smoothness controls the usual bias–variance tradeoff: stronger smoothing generally reduces variance and can increase bias.

9.4 Local Regression

Local regression is another way to fit a flexible nonlinear function. At a target value x_0, use nearby training observations and fit a low-degree polynomial with weights decreasing with distance from x_0. A local linear fit solves

$$\min_{\beta_0, \beta_1} \sum_{i=1}^{n} w_i(x_0)(y_i - \beta_0 - \beta_1 x_i)^2,$$

and predicts

$$\widehat{f}(x_0) = \widehat{\beta}_0 + \widehat{\beta}_1 x_0.$$

Equivalently, using $x_i - x_0$ in the polynomial makes its fitted intercept the prediction at x_0.

The procedure requires three choices: a span, a weighting kernel, and a polynomial degree. A span can specify the fraction of observations used around x_0; a larger span uses a broader neighborhood. With a compactly supported kernel, observations outside the span have zero weight. A Gaussian kernel is another possible choice,

$$w_i(x_0) \propto \exp\left[-\frac{(x_i - x_0)^2}{2h^2}\right],$$

though its weights are strictly positive at all finite distances unless explicitly truncated. In either case the nearest observations receive the greatest weights.

The span or bandwidth is the main smoothness tuning parameter. Small neighborhoods allow more variation and can increase variance; large neighborhoods smooth more and can increase bias. As in K-nearest-neighbor regression, the fit is computed locally rather than summarized by one global coefficient vector. The training observations are needed again to make predictions at new target values.

9.5 Generalized Additive Models

The preceding methods allow flexible fitting with one predictor. For several predictors, an additive regression model uses a separate function for each predictor:

$$\mathbb{E}(Y \mid X = x) = f(x) = \beta_0 + f_1(x_1) + \cdots + f_p(x_p).$$

Each f_j captures the effect of x_j on the response and serves as a building block. The model extends a linear regression by replacing $\beta_j x_j$ with $f_j(x_j)$ and adding the separate effects. Centering constraints, such as $\sum_i f_j(x_{ij}) = 0$, distinguish each function from the intercept.

A generalized additive model uses the same additive predictor through a link. For binary data,

$$\log \frac{p(x)}{1 - p(x)} = \beta_0 + \sum_{j=1}^{p} f_j(x_j).$$

Splines or local smoothers can supply the component functions. Additivity permits nonlinear effects in individual predictors while keeping their contributions separately interpretable.

Chapter 10

Tree-Based Models

10.1 Regression Trees and Terminal Regions

A tree partitions predictor space into nonoverlapping regions. Each terminal node, or leaf, corresponds to a region and supplies a prediction for observations falling there. Standard classification and regression trees use axis-aligned splits, producing rectangular boxes in two dimensions and their higher-dimensional analogues.

For a regression tree with terminal regions $R_1, \dots, R_J$, the fitted mean function is piecewise constant:

$$f(x) = \sum_{j=1}^{J} c_j \mathbf{1}\{x \in R_j\}.$$

To fit the tree by least squares, consider

$$\begin{aligned} \mathrm{RSS} &= \sum_{i=1}^{n} \big(y_i - f(x_i)\big)^2 \\ &= \sum_{j=1}^{J} \sum_{i : x_i \in R_j} (y_i - c_j)^2 = \sum_{j=1}^{J} n_j Q_j, \end{aligned}$$

where

$$n_j = \#\{i : x_i \in R_j\}, \qquad Q_j = \frac{1}{n_j} \sum_{i : x_i \in R_j} (y_i - c_j)^2.$$

The quantity Q_j measures node impurity. At its minimizing constant it is the node's mean squared error.

Exercise 10.1. For a fixed nonempty region R_j, find the constant that minimizes its contribution to RSS.

Solution. Let $\bar{y}_j = n_j^{-1} \sum_{i : x_i \in R_j} y_i$. Since $\sum_{i : x_i \in R_j} (y_i - \bar{y}_j) = 0$,

$$\sum_{i : x_i \in R_j} (y_i - c)^2 = \sum_{i : x_i \in R_j} (y_i - \bar{y}_j)^2 + n_j (c - \bar{y}_j)^2.$$

The first term is independent of c and the second is minimized uniquely at $c = \bar{y}_j$. Thus

$$\widehat{c}_j = \mathrm{ave}\{y_i : x_i \in R_j\}.$$

Optimizing over all possible region shapes and all constants is not computationally feasible. Recursive binary splitting restricts the search to successive binary, axis-aligned splits. It is top-down: begin with a single region containing all observations. At each step choose the best split of a current region. It is greedy: the split optimizes the present step rather than looking ahead over all possible later trees.

10.2 Recursive Binary Splitting

For a splitting variable m and cutpoint s, define

$$R_1(m,s) = \{x : x_m < s\}, \qquad R_2(m,s) = \{x : x_m \geq s\}.$$

Assigning equality to the second region makes these sets a partition. When splitting a region below the root, intersect each set with that parent region.

For a candidate split, choose c_1, c_2 to minimize

$$\sum_{i:x_i \in R_1(m,s)} (y_i - c_1)^2 + \sum_{i:x_i \in R_2(m,s)} (y_i - c_2)^2.$$

For any (m,s) producing nonempty children, the minimizing values are

$$\widehat{c}_1(m,s) = \text{ave}\{y_i : x_i \in R_1(m,s)\}, \qquad \widehat{c}_2(m,s) = \text{ave}\{y_i : x_i \in R_2(m,s)\}.$$

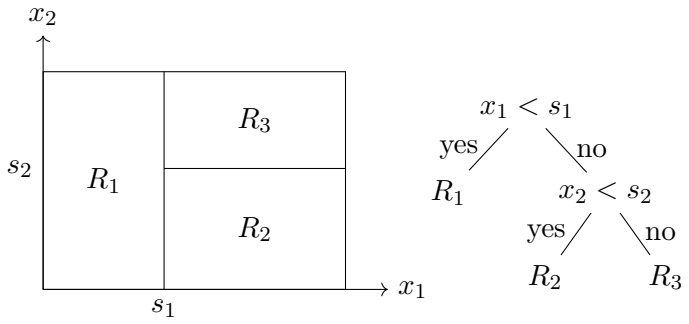
The split search therefore solves

$$\min_{m,s} \left\{ \sum_{i:x_i \in R_1(m,s)} (y_i - \widehat{c}_1(m,s))^2 + \sum_{i:x_i \in R_2(m,s)} (y_i - \widehat{c}_2(m,s))^2 \right\}.$$

Only changes in which observations fall on either side matter, so cutpoints between successive distinct observed values suffice. Repeat the splitting procedure until the stopping criterion is reached. A minimum node size or a limit on splitting can prevent terminal regions from becoming too small.

A new observation receives the average response among training observations in its terminal region:

$$\widehat{f}(x) = \widehat{c}_j \quad \text{when } x \in R_j.$$



The partition and the tree describe the same successive splits. A fitted tree can become too complex: too many splits fit training fluctuations and cause overfitting. Bagging, random forests, and boosting are related tree-based approaches that combine trees rather than relying on a single fitted tree.

10.3 Classification Trees

The same partition idea applies to a categorical response. A terminal region contains a fitted distribution of class labels rather than a fitted numerical mean.

Exercise 10.2. For a terminal region containing n_j observations from classes $1, \dots, K$, find the fitted class proportions and the class prediction minimizing the region's training misclassification rate.

Solution. If n_{jk} observations in the region belong to class k, the empirical class proportion is

$$\widehat{p}_{jk} = \frac{n_{jk}}{n_j}.$$

Predicting class k misclassifies $n_j - n_{jk}$ observations, a proportion $1 - \widehat{p}_{jk}$. Therefore any class attaining

$$\widehat{k}_j \in \operatorname*{arg\,max}_{1 \le k \le K} \widehat{p}_{jk}$$

minimizes this rate, whose minimum is $1 - \max_k \widehat{p}_{jk}$. Ties may be resolved by a fixed rule. New observations in the region receive this majority-class prediction.

Chapter 11

Maximum-Margin Classification

11.1 Separating Hyperplanes and the Margin

Let $x_i \in \mathbb{R}^p$ and $y_i \in \{-1, +1\}$ for $i = 1, \dots, n$, with both classes represented. Assume the training data are strictly linearly separable: there exist β_0 and a nonzero vector β such that

$$y_i(\beta_0 + x_i^T \beta) > 0 \quad \text{for every } i.$$

Define

$$f(x) = \beta_0 + x^T \beta, \qquad \|\beta\|_2 = \sqrt{\beta^T \beta}.$$

The decision boundary is the hyperplane $f(x) = 0$; the classifier predicts the sign of $f(x)$, with a convention for points exactly on the boundary.

The perpendicular distance from x_i to this hyperplane is

$$\frac{|f(x_i)|}{\|\beta\|_2}.$$

For a separating hyperplane, $y_i f(x_i) = |f(x_i)|$. Its margin, the shortest distance from any training observation to the boundary, is

$$M(\beta_0, \beta) = \frac{1}{\|\beta\|_2} \min_{1 \le i \le n} |f(x_i)| = \min_i \frac{y_i f(x_i)}{\|\beta\|_2} > 0.$$

The maximum-margin classifier selects the separating hyperplane with largest margin.

Exercise 11.1. Verify the distance formula and describe the empty slab associated with a separating hyperplane of margin M.

Solution. The vector β is normal to the hyperplane. The projection of x_i onto it is

$$x_i^* = x_i - \frac{f(x_i)}{\|\beta\|_2^2} \beta,$$

because

$$f(x_i^*) = f(x_i) - \frac{f(x_i)}{\|\beta\|_2^2} \beta^T \beta = 0.$$

Consequently

$$\|x_i - x_i^*\|_2 = \frac{|f(x_i)|}{\|\beta\|_2^2} \|\beta\|_2 = \frac{|f(x_i)|}{\|\beta\|_2}.$$

The margin inequalities are $y_i f(x_i) \geq M \|\beta\|_2$. Therefore the open slab

$$-M \|\beta\|_2 < f(x) < M \|\beta\|_2$$

contains no training observations. Its boundary hyperplanes are $f(x) = \pm M \|\beta\|_2$. Each lies distance M from the central hyperplane, and their separation is $2M$.

11.2 Equivalent Optimization Formulations

The geometric optimization can be written

$$\max_{\beta_0,\, \beta \neq 0,\, M} M \quad \text{subject to} \quad y_i(\beta_0 + x_i^T \beta) \geq M \|\beta\|_2, \quad i = 1, \ldots, n.$$

Multiplying both β_0 and β by the same positive constant leaves the hyperplane and geometric margin unchanged. In particular, $f(x) = 0$ and $f(x)/\|\beta\|_2 = 0$ define the same boundary. We may exploit this scale freedom in two ways.

First, impose $\|\beta\|_2 = 1$ and solve

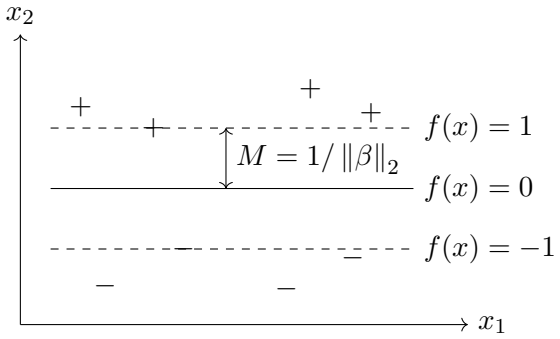
$$\max_{\beta_0, \beta, M} M \quad \text{subject to} \quad \|\beta\|_2 = 1, \quad y_i(\beta_0 + x_i^T \beta) \geq M.$$

Second, rescale the coefficients so that the smallest functional margin is one. Then $M \|\beta\|_2 = 1$ and maximizing $M = 1/\|\beta\|_2$ is equivalent to

$$\min_{\beta_0, \beta} \frac{1}{2} \|\beta\|_2^2 \quad \text{subject to} \quad y_i(\beta_0 + x_i^T \beta) \geq 1, \quad i = 1, \ldots, n.$$

This is the hard-margin optimization problem. Strict separability guarantees feasibility. At an optimum the smallest functional margin is exactly one: if it were greater than one, rescaling both coefficients downward would preserve feasibility while reducing the objective.

In this normalization the slab boundaries are $f(x) = +1$ and $f(x) = -1$. The geometric margin is $1/\|\beta\|_2$ and the full slab width is $2/\|\beta\|_2$.



11.3 Lagrange Multipliers and Support Vectors

Associate a multiplier $\alpha_i \geq 0$ with the constraint $y_i f(x_i) - 1 \geq 0$. The Lagrangian is

$$L(\beta_0, \beta, \alpha) = \frac{1}{2} \|\beta\|_2^2 - \sum_{i=1}^{n} \alpha_i \big[y_i(\beta_0 + x_i^T \beta) - 1\big].$$

Exercise 11.2. Derive the stationarity and complementary-slackness equations. Explain which observations determine the slope vector and how to recover the intercept.

Solution. Stationarity with respect to the intercept gives

$$\frac{\partial L}{\partial \beta_0} = -\sum_{i=1}^{n} \alpha_i y_i = 0.$$

Stationarity with respect to the slope vector gives

$$\nabla_\beta L = \beta - \sum_{i=1}^{n} \alpha_i y_i x_i = 0, \qquad \beta = \sum_{i=1}^{n} \alpha_i y_i x_i.$$

Together with primal and dual feasibility, the optimality conditions are

$$\begin{aligned} y_i f(x_i) &\geq 1, & \alpha_i &\geq 0, \\ \sum_{i=1}^{n} \alpha_i y_i &= 0, & \beta &= \sum_{i=1}^{n} \alpha_i y_i x_i, \\ \alpha_i \big[y_i f(x_i) - 1\big] &= 0 & &\text{for each } i. \end{aligned}$$

Strict separability gives a strictly feasible point after a sufficient positive rescaling, so these conditions characterize the optimum of the convex hard-margin problem.

If $y_i f(x_i) > 1$, complementary slackness forces $\alpha_i = 0$. Hence observations strictly outside the slab boundaries contribute nothing to the slope representation. Observations with $\alpha_i > 0$ lie on a slab boundary and are support vectors. In a degenerate configuration, a point can lie on a boundary with zero multiplier; it then does not contribute to this representation.

For any boundary observation satisfying $y_i f(x_i) = 1$, use $y_i^2 = 1$ to obtain

$$\beta_0 + x_i^T \beta = y_i, \qquad \beta_0 = y_i - x_i^T \beta.$$

The classifier is consequently determined by the supporting observations and their multipliers, with the intercept recovered from an active boundary constraint.

Chapter 12

Clustering

12.1 Unsupervised Learning and Dissimilarity

Unsupervised learning explores structure in predictors without observed response labels. PCA describes directions of variation; clustering seeks groups of observations. In classification the response classes are given in advance. In clustering we instead ask whether the observations form meaningful groups.

In K-means clustering, the number K is specified before fitting. Hierarchical clustering constructs a nested collection of groups summarized by a tree, called a dendrogram, so a final number of clusters need not be chosen at the outset.

Let $x_i = (x_{i1}, \dots, x_{ip})^T$ denote observation i. For K-means the dissimilarity is squared Euclidean distance,

$$d_2^2(x_i, x_\ell) = \sum_{j=1}^{p} (x_{ij} - x_{\ell j})^2 = \|x_i - x_\ell\|_2^2 .$$

A partition $C = (C_1, \dots, C_K)$ consists of nonempty, pairwise disjoint sets of observation indices whose union is $\{1, \dots, n\}$. Thus every observation belongs to exactly one cluster. Let

$$n_k = |C_k|, \qquad \bar{x}_k = \frac{1}{n_k} \sum_{i \in C_k} x_i$$

be its size and centroid.

12.2 Within- and Between-Cluster Sums of Squares

Define the within-cluster sum of squares by

$$W(C_k) = \frac{1}{2n_k} \sum_{i \in C_k} \sum_{\ell \in C_k} \|x_i - x_\ell\|_2^2 = \sum_{i \in C_k} \|x_i - \bar{x}_k\|_2^2 .$$

The double sum is over ordered pairs; the factor $1/(2n_k)$ is important. The total within-cluster sum of squares is

$$W(C) = \sum_{k=1}^{K} W(C_k).$$

It measures overall dissimilarity within the chosen groups.

Let $\bar{x} = n^{-1} \sum_i x_i$. The total sum of squares is

$$T = \sum_{i=1}^{n} \|x_i - \bar{x}\|_2^2 ,$$

which does not depend on the partition. The between-cluster sum of squares is

$$B(C) = \sum_{k=1}^{K} n_k \, \|\bar{x}_k - \bar{x}\|_2^2 .$$

Exercise 12.1. Prove the pairwise-distance identity for $W(C_k)$ and the decomposition $T = W(C) + B(C)$. Deduce the equivalent optimization criteria for a fixed number of clusters.

Solution. For a cluster with $m = n_k$ observations and centroid $\mu = \bar{x}_k$,

$$\begin{aligned}
\sum_{i \in C_k} \sum_{\ell \in C_k} \|x_i - x_\ell\|_2^2 &= 2m \sum_{i \in C_k} \|x_i\|_2^2 - 2 \left\| \sum_{i \in C_k} x_i \right\|_2^2 \\
&= 2m \left(\sum_{i \in C_k} \|x_i\|_2^2 - m \, \|\mu\|_2^2 \right) \\
&= 2m \sum_{i \in C_k} \|x_i - \mu\|_2^2 .
\end{aligned}$$

Dividing by $2m$ proves the first identity.

Next write $x_i - \bar{x} = (x_i - \bar{x}_k) + (\bar{x}_k - \bar{x})$ for $i \in C_k$. Expanding the squared norm gives

$$\begin{aligned}
\sum_{i \in C_k} \|x_i - \bar{x}\|_2^2 = \sum_{i \in C_k} \|x_i - \bar{x}_k\|_2^2 + n_k \, \|\bar{x}_k - \bar{x}\|_2^2 \\
+ 2 \left(\sum_{i \in C_k} (x_i - \bar{x}_k) \right)^T (\bar{x}_k - \bar{x}).
\end{aligned}$$

The last term is zero by the definition of the centroid. Summing over clusters gives

$$T = W(C) + B(C), \qquad W(C) = T - B(C).$$

Since T is fixed, minimizing $W(C)$ over partitions into K nonempty clusters is equivalent to maximizing $B(C)$ over those partitions.

12.3 The K-Means Algorithm

The K-means objective is

$$\min_{C_1, \dots, C_K} \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \bar{x}_k\|_2^2 .$$

It can also be written with both assignments and centers unknown:

$$\min_{C,\, m_1, \dots, m_K} Q(C, m), \qquad Q(C, m) = \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - m_k\|_2^2 .$$

For fixed assignments, the minimizing center is $m_k = \bar{x}_k$. Exhaustively searching all partitions is difficult, so the usual algorithm uses iterative descent.

Exercise 12.2. Starting from an initial cluster assignment, give the two alternating steps of the K-means algorithm and show that neither step increases the objective.

Solution. Start with a partition $C^{(0)}$ into K nonempty clusters. Repeat the following steps.

1. Compute the centroid of each current cluster:

$$m_k^{(t)} = \frac{1}{|C_k^{(t)}|} \sum_{i \in C_k^{(t)}} x_i.$$

2. Assign each observation to a nearest centroid:

$$c_i^{(t+1)} \in \arg\min_{1 \le k \le K} \left\| x_i - m_k^{(t)} \right\|_2^2, \qquad C_k^{(t+1)} = \{i : c_i^{(t+1)} = k\}.$$

Use a fixed tie rule. If a cluster becomes empty, the algorithm must specify how to handle it, for example by moving its unused center to an observation before reassignment; the monotonicity statements below concern the nonempty centroid updates and the nearest-center assignments themselves.

For a fixed cluster C_k and any center m,

$$\sum_{i \in C_k} \|x_i - m\|_2^2 = \sum_{i \in C_k} \|x_i - \bar{x}_k\|_2^2 + n_k \|m - \bar{x}_k\|_2^2 .$$

Thus replacing a center by the cluster centroid cannot increase Q. For fixed centers, each observation's nearest-center assignment minimizes its individual squared-distance contribution. Summing those minimized contributions shows that reassignment cannot increase Q either. Consequently the successive objective values are nonincreasing and bounded below by zero.

Stop when assignments cease changing, or when improvement is sufficiently small. At a fixed point, each center is a centroid and each observation is assigned to a nearest center. This is an alternating-minimization solution; the procedure is not guaranteed to find the globally best partition.

K must be specified in advance, and the result is sensitive to the initial assignment. Different starting partitions can reach different solutions. Comparing several initializations by their final within-cluster sum of squares reduces this dependence but does not guarantee a global optimum.

12.4 Agglomerative Hierarchical Clustering

Hierarchical clustering does not require a single chosen K before fitting. The bottom-up, or agglomerative, procedure begins with one cluster per observation and repeatedly merges two clusters. There are initially n clusters, then $n-1$, then $n-2$, and eventually one root cluster. Exactly $n-1$ merges are required.

The algorithm is:

1. Compute the pairwise observation dissimilarities, initially $\binom{n}{2}$ values, and begin with the n singleton clusters.
2. Find the two current clusters with smallest intercluster dissimilarity and merge them.
3. Recompute the dissimilarities between the newly merged cluster and the remaining clusters using the chosen linkage rule; keep the other dissimilarities unchanged.
4. Repeat until all observations belong to one cluster.

The linkage rule specifies how observation dissimilarities become a dissimilarity between clusters.

12.4.1 Linkage Rules

Let A and B be disjoint nonempty clusters, and let $d(x_i, x_\ell)$ be the chosen observation dissimilarity. Common rules are

$$D_{\text{single}}(A, B) = \min_{i \in A,\, \ell \in B} d(x_i, x_\ell),$$
$$D_{\text{complete}}(A, B) = \max_{i \in A,\, \ell \in B} d(x_i, x_\ell),$$
$$D_{\text{average}}(A, B) = \frac{1}{|A||B|} \sum_{i \in A} \sum_{\ell \in B} d(x_i, x_\ell).$$

Single linkage uses nearest neighbors across the two clusters. Complete linkage uses farthest neighbors. Average linkage uses the average of all cross-cluster dissimilarities. For Euclidean observations, centroid linkage uses

$$D_{\text{centroid}}(A, B) = \|\bar{x}_A - \bar{x}_B\|_2 .$$

Single linkage can create long, trailing clusters through *chaining*: a short connection between two groups is enough for them to merge even when their overall diameter is large. Its clusters need not be compact. Complete linkage favors compact clusters by controlling the largest cross-cluster distance, but distinct resulting groups can still have members lying close to one another. Average linkage compromises between these extremes.

Desirable groupings have members close to one another and groups well separated from one another. It can also be desirable for a clustering method to preserve the grouping under strictly increasing transformations of the supplied dissimilarities. Single and complete linkage have this invariance, with tie decisions handled consistently; average linkage does not have it in general.

Exercise 12.3. Explain the transformation invariance of single and complete linkage and why it need not hold for average linkage.

Solution. If h is strictly increasing, then for any finite set of distances,

$$\min h(d) = h(\min d), \qquad \max h(d) = h(\max d).$$

Thus every single- or complete-linkage dissimilarity is transformed by the same increasing function h. Its ordering among cluster pairs is unchanged. Starting with the singleton clusters and arguing at each merge, the entire merge sequence is unchanged, provided the same tie rule is used. Merge heights are transformed, but the nested groups are preserved.

For average linkage, in general

$$\frac{1}{|A||B|}\sum_{i\in A,\ell\in B} h(d_{i\ell}) \neq h\left(\frac{1}{|A||B|}\sum_{i\in A,\ell\in B} d_{i\ell}\right).$$

For example, at a current stage let $A = \{a_1, a_2\}$, $B = \{b\}$, and $C = \{c\}$, with cross-cluster dissimilarities

$$d(a_1, b) = 1, \quad d(a_2, b) = 5, \quad d(a_1, c) = d(a_2, c) = 3.1, \quad d(b, c) = 4.$$

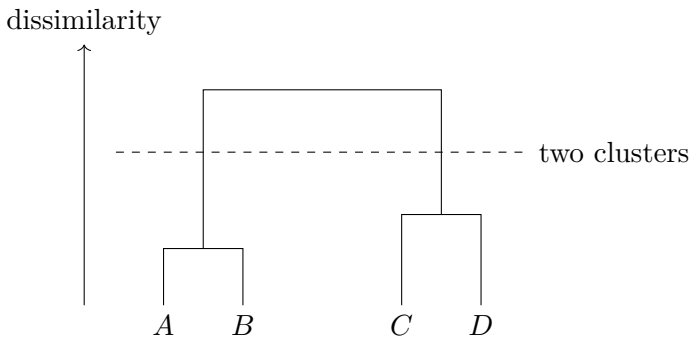
The average-linkage distances are $D(A, B) = 3$, $D(A, C) = 3.1$, and $D(B, C) = 4$, so A and B are preferred. After squaring each dissimilarity, these become 13, 9.61, and 16, so A and C are preferred. Averaging and a nonlinear increasing transformation can change the next merge. A common positive scaling, however, commutes with averaging and preserves its ordering.

Centroid linkage can produce an *inversion*: after a merge, the new centroid can be closer to another cluster than the previous merge height. For example, three points at the vertices of an equilateral triangle of side length two have pairwise distances two. Merging any two places their centroid at the midpoint of one side. Its distance to the remaining vertex is $\sqrt{3} < 2$, so the next merge occurs at a smaller dissimilarity. This complicates interpreting merge heights as increasing levels of separation.

12.4.2 Interpreting a Dendrogram

A dendrogram records the merges and their dissimilarities. Its vertical axis shows the dissimilarity at which clusters merge. A lower merge height indicates greater similarity under the selected measure and linkage. Cutting the tree horizontally gives the groups existing at that level; for a hierarchy with nondecreasing heights, one can choose a cut to obtain the desired number of clusters, subject to ties.

Horizontal proximity between leaves is not a measure of dissimilarity. Branches can be swapped around an internal node without changing any merge or its height.



A hierarchical structure is a modeling restriction. Real data may not have meaningful nested groups, so a dendrogram does not by itself establish that such a structure exists.

12.5 Choosing a Dissimilarity Measure

12.5.1 Metric-Based Measures

The Minkowski distance, for $q \geq 1$, is

$$d_q(x_i, x_\ell) = \left(\sum_{j=1}^{p} |x_{ij} - x_{\ell j}|^q \right)^{1/q}.$$

For $q = 1$ it is Manhattan distance; for $q = 2$ it is Euclidean distance. The choice determines what kinds of differences between observations are treated as large. Squared Euclidean dissimilarity is useful for the K-means sum-of-squares objective, although it is not itself a metric because it need not satisfy the triangle inequality.

12.5.2 Correlation-Based Measures

Sometimes the pattern across variables matters more than the overall magnitude of an observation. Let

$$\bar{x}_{i\cdot} = \frac{1}{p} \sum_{j=1}^{p} x_{ij}.$$

For two rows with nonzero variation across their coordinates, define their correlation by

$$r(x_i, x_\ell) = \frac{\sum_{j=1}^{p} (x_{ij} - \bar{x}_{i\cdot})(x_{\ell j} - \bar{x}_{\ell\cdot})}{\sqrt{\sum_{j=1}^{p} (x_{ij} - \bar{x}_{i\cdot})^2} \sqrt{\sum_{j=1}^{p} (x_{\ell j} - \bar{x}_{\ell\cdot})^2}}.$$

The coordinates, rather than the observations, index this correlation calculation. A common dissimilarity is

$$d_{\mathrm{corr}}(x_i, x_\ell) = 1 - r(x_i, x_\ell).$$

This is not generally a metric. For instance, unit vectors in a two-dimensional plane separated successively by 60 degrees have neighboring dissimilarities $1 - \cos 60^\circ = 1/2$, while the two extreme vectors have dissimilarity $1 - \cos 120^\circ = 3/2 > 1$. Such a plane can be chosen within the centered-row subspace when $p \geq 3$, so the same triangle-inequality failure occurs for centered row correlations.

Exercise 12.4. Relate correlation dissimilarity to squared Euclidean distance after row standardization. State when the resulting hierarchical clusterings agree.

Solution. Center and normalize each nonconstant row to unit Euclidean norm:

$$z_i = \frac{x_i - \bar{x}_{i\cdot}\mathbf{1}}{\|x_i - \bar{x}_{i\cdot}\mathbf{1}\|_2}.$$

Then $z_i^T z_\ell = r(x_i, x_\ell)$ and $\|z_i\|_2 = 1$, so

$$\|z_i - z_\ell\|_2^2 = \|z_i\|_2^2 + \|z_\ell\|_2^2 - 2 z_i^T z_\ell = 2\big(1 - r(x_i, x_\ell)\big).$$

If instead each row is divided by its sample standard deviation across its p coordinates,

$$s_i^2 = \frac{1}{p-1} \sum_{j=1}^{p} (x_{ij} - \bar{x}_{i\cdot})^2, \qquad u_{ij} = \frac{x_{ij} - \bar{x}_{i\cdot}}{s_i},$$

then $\|u_i\|_2^2 = p - 1$ and

$$\|u_i - u_\ell\|_2^2 = 2(p-1)(1 - r(x_i, x_\ell)).$$

These are row centering and scaling operations. Standardizing each predictor column across observations does not in general give these identities.

Thus squared Euclidean dissimilarity on these standardized rows is a common positive multiple of correlation dissimilarity. For single, complete, and average linkage, that common scaling preserves all comparisons and the merge sequence, with consistent tie handling. Using Euclidean distance itself takes an additional square root. Single and complete linkage preserve the merge sequence under this strictly increasing transformation, but average linkage need not. Therefore correlation-based and Euclidean-based dendrograms are equivalent only with the appropriate standardization and linkage qualification; their numerical merge heights also use different scales.